XIAMEN UNIVERSITY MALAYSIA

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Feedback from Lecturer:

Mark:

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1.0 Introduction

1.1 Dataset Description

This dataset was collected from Kaggle. The link is: https://www.kaggle.com/datasets/valakhorasani/gym-members-exercise-dataset

This dataset provides a detailed overview of gym members' exercise routines, physical attributes, and fitness metrics. It contains 973 samples of gym data, including key performance indicators such as heart rate, calories burned, and workout duration. Each entry also includes demographic data and experience levels, allowing for comprehensive analysis of fitness patterns, athlete progression, and health trends.

Key Features:

- Age: Age of the gym member.
- Gender: Gender of the gym member (Male or Female).
- Weight (kg): Member’s weight in kilograms.
- Height (m): Member’s height in meters.
- Max_BPM: Maximum heart rate (beats per minute) during workout sessions.
- Avg_BPM: Average heart rate during workout sessions.
- Resting_BPM: Heart rate at rest before workout.
- Session_Duration (hours): Duration of each workout session in hours.
- Calories_Burned: Total calories burned during each session.
- Workout_Type: Type of workout performed (e.g., Cardio, Strength, Yoga, HIIT).
- Fat_Percentage: Body fat percentage of the member.
- Water_Intake (liters): Daily water intake during workouts.
- Workout_Frequency (days/week): Number of workout sessions per week.
- Experience_Level: Level of experience, from beginner (1) to expert (3).
- BMI: Body Mass Index, calculated from height and weight.

1.2 Data Cleaning and Preprocessing

Before performing any analysis, we conducted a series of **data cleaning and preprocessing** steps:

```
```{r}
path <- "C:/University/Year 3 Sem 1/Statistical Programming using R/Assignment/gym_members_exercise_tracking.csv"
data <- read.csv(path, header = TRUE, sep = ",")
attach(data)

#preprocessing
sum(is.na(data))
anyNA(data)

str(data)
head(data)
summary(data)

```
```

The dataset was loaded using the read.csv() function,performed a check for missing data, and see the structure of the data:

```
[1] 0
[1] FALSE
'data.frame':	973 obs. of  15 variables:
 $ Age                          : int  56 46 32 25 38 56 36 40 28 28 ...
 $ Gender                       : chr  "Male" "Female" "Female" "Male" ...
 $ Weight..kg.                  : num  88.3 74.9 68.1 53.2 46.1 ...
 $ Height..m.                   : num  1.71 1.53 1.66 1.7 1.79 1.68 1.72 1.51 1.94
1.84 ...
 $ Max_BPM                      : int  180 179 167 190 188 168 174 189 185 169 ...
 $ Avg_BPM                      : int  157 151 122 164 158 156 169 141 127 136 ...
 $ Resting_BPM                  : int  60 66 54 56 68 74 73 64 52 64 ...
 $ Session_Duration..hours.     : num  1.69 1.3 1.11 0.59 0.64 1.59 1.49 1.27 1.03
1.08 ...
 $ Calories_Burned              : int  1313 883 677 532 556 1116 1385 895 719 808 ..
 $ Workout_Type                 : chr  "Yoga" "HIIT" "Cardio" "Strength" ...
 $ Fat_Percentage               : num  12.6 33.9 33.4 28.8 29.2 15.5 21.3 30.6 28.9
29.7 ...
 $ Water_Intake..liters.        : num  3.5 2.1 2.3 2.1 2.8 2.7 2.3 1.9 2.6 2.7 ...
 $ Workout_Frequency..days.week.: int  4 4 4 3 3 5 3 3 4 3 ...
 $ Experience_Level             : int  3 2 2 1 1 3 2 2 2 1 ...
 $ BMI                          : num  30.2 32 24.7 18.4 14.4 ...
```

	Age <int>	Gender <chr>	Weight..kg. <dbl>	Height..m. <dbl>	Max_BPM <int>	Avg_BPM <int>	Resting_BPM <int>	Session_Duration..hours. <dbl>	Calories_Burned <int>
1	56	Male	88.3	1.71	180	157	60	1.69	1313
2	46	Female	74.9	1.53	179	151	66	1.30	883
3	32	Female	68.1	1.66	167	122	54	1.11	677
4	25	Male	53.2	1.70	190	164	56	0.59	532
5	38	Male	46.1	1.79	188	158	68	0.64	556
6	56	Female	58.0	1.68	168	156	74	1.59	1116

6 rows | 1-10 of 15 columns

Calories_Burned <int>	Workout_Type <chr>	Fat_Percentage <dbl>	Water_Intake..liters. <dbl>	Workout_Frequency..days.week. <int>	Experience_Level <int>	BMI <dbl>
1313	Yoga	12.6	3.5	4	3	30.20
883	HIIT	33.9	2.1	4	2	32.00
677	Cardio	33.4	2.3	4	2	24.71
532	Strength	28.8	2.1	3	1	18.41
556	Strength	29.2	2.8	3	1	14.39
1116	HIIT	15.5	2.7	5	3	20.55

6 rows | 10-16 of 15 columns

Then, do some data cleaning:

```
```{r}
#transform Experience_Level to factor
data$Experience_Level <- factor(data$Experience_Level, levels = 1:3, labels = c("Beginner", "Intermediate", "Expert"))

#install.packages("janitor")
library(janitor)
data <- janitor::clean_names(data)
names(data)
str(data)
```
```

```
 [1] "age"                          "gender"                       "weight_kg"                    "height_m"                     "max_bpm"
 [6] "avg_bpm"                      "resting_bpm"                  "session_duration_hours"       "calories_burned"              "workout_type"
[11] "fat_percentage"               "water_intake_liters"          "workout_frequency_days_week"  "experience_level"             "bmi"
'data.frame':	973 obs. of  15 variables:
 $ age                        : int  56 46 32 25 38 56 36 40 28 28 ...
 $ gender                     : chr  "Male" "Female" "Female" "Male" ...
 $ weight_kg                  : num  88.3 74.9 68.1 53.2 46.1 ...
 $ height_m                   : num  1.71 1.53 1.66 1.7 1.79 1.68 1.72 1.51 1.94 1.84 ...
 $ max_bpm                    : int  180 179 167 190 188 168 174 189 185 169 ...
 $ avg_bpm                    : int  157 151 122 164 158 156 169 141 127 136 ...
 $ resting_bpm                : int  60 66 54 56 68 74 73 64 52 64 ...
 $ session_duration_hours     : num  1.69 1.3 1.11 0.59 0.64 1.59 1.49 1.27 1.03 1.08 ...
 $ calories_burned            : int  1313 883 677 532 556 1116 1385 895 719 808 ...
 $ workout_type               : chr  "Yoga" "HIIT" "Cardio" "Strength" ...
 $ fat_percentage             : num  12.6 33.9 33.4 28.8 29.2 15.5 21.3 30.6 28.9 29.7 ...
 $ water_intake_liters        : num  3.5 2.1 2.3 2.1 2.8 2.7 2.3 1.9 2.6 2.7 ...
 $ workout_frequency_days_week: int  4 4 4 3 3 5 3 3 4 3 ...
 $ experience_level           : Factor w/ 3 levels "Beginner","Intermediate",..: 3 2 2 1 1 3 2 2 2 1 ...
 $ bmi                        : num  30.2 32 24.7 18.4 14.4 ...
```

The Experience_Level variable was originally encoded as numeric values (1 to 3). For meaningful analysis, it was converted into a **categorical factor** (“Beginner”, “Intermediate”, “Expert”). This step improves interpretability and allows proper treatment in statistical models that recognize factor levels.

To enhance consistency and avoid syntax issues in analysis, the clean_names() function from the janitor package was used. This function converts variable names to a **snake_case** format, for example, “Workout_Frequency (days/week)” become “workout_frequency_days_week”, making them easier to reference in R code and reducing the chance of syntax errors. Now, the dataset is prepoccessed.

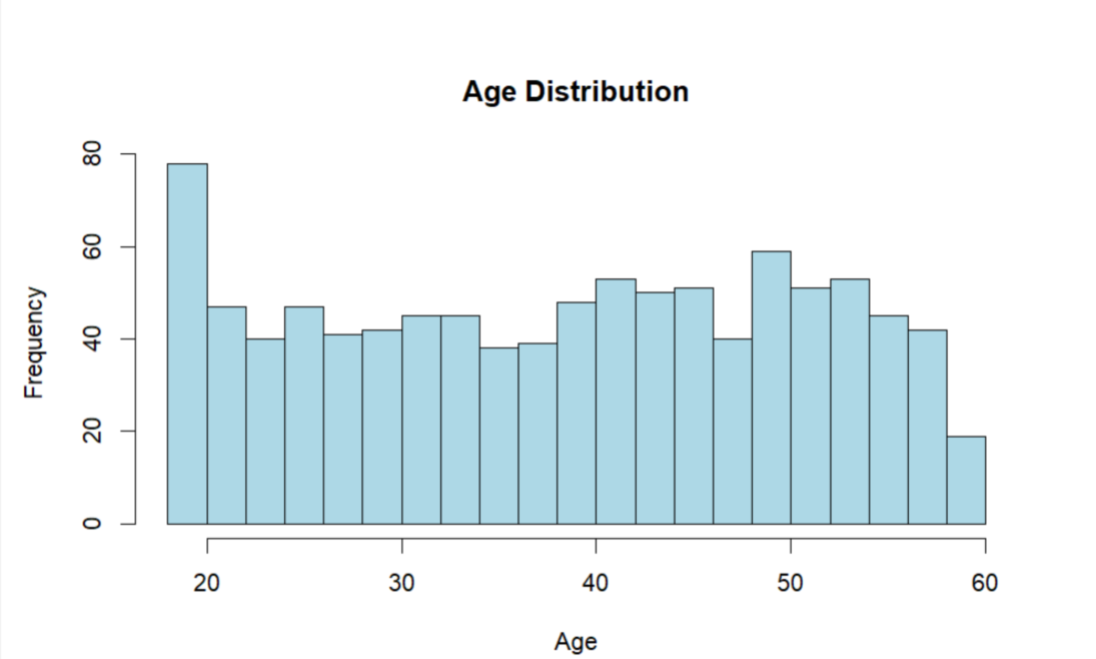
2.0 Exploratory Data Analysis (EDA):

2.1 Data Visualization

Some exploratory data analysis has been performed to get a better understanding to the data set.

First, the distribution of age has been performed using a histogram, it is because age is a continuous numerical variable.

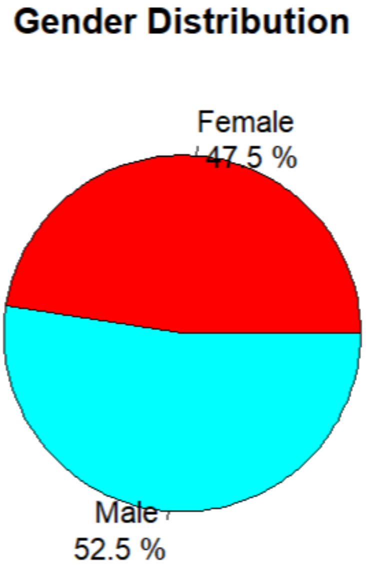
```{r}
#Basic data analysis to understand the dataset
#2.1 Age distribution
hist(data$age,
     main = "Age Distribution",
     xlab = "Age",
     col = "lightblue",
     border = "black",
     breaks = 20)
```

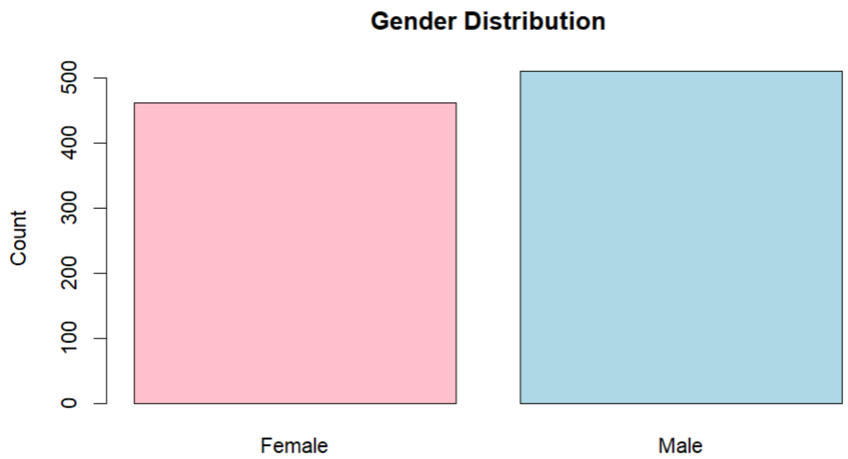


The member's age is quite evenly distributed among 20 to 60 years old. There are slightly more members who are slightly younger than 20 years old compared to others.

```{r}
#gender proportion
gender_table <- table(data$gender)
gender_table
barplot(gender_table,
        main = "Gender Distribution",
        col = c("pink", "lightblue"),
        ylab = "Count")


pie(gender_table,
    main = "Gender Distribution",
    col = rainbow(length(gender_table)),
    labels = paste(names(gender_table), "\n", round(100 * gender_table/sum(gender_table), 1), "%"))
```

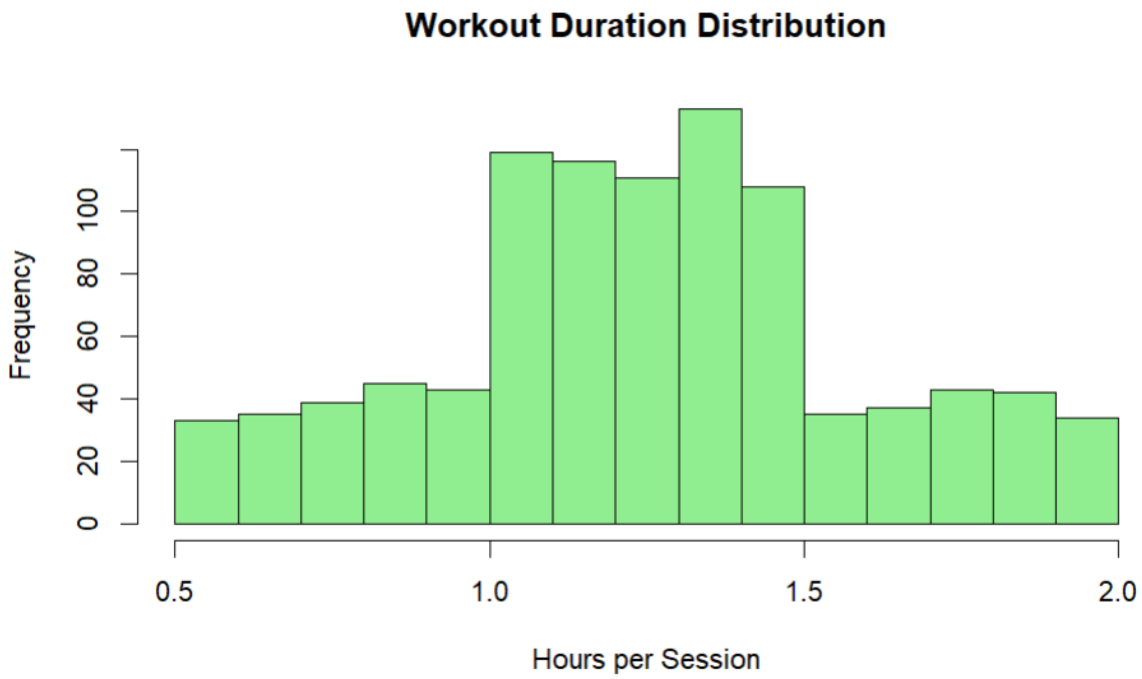
```
Female   Male
   462    511
```





The graph shows that 52.5% of members (511 members) are male while 47.5% of members are female (462 members) .

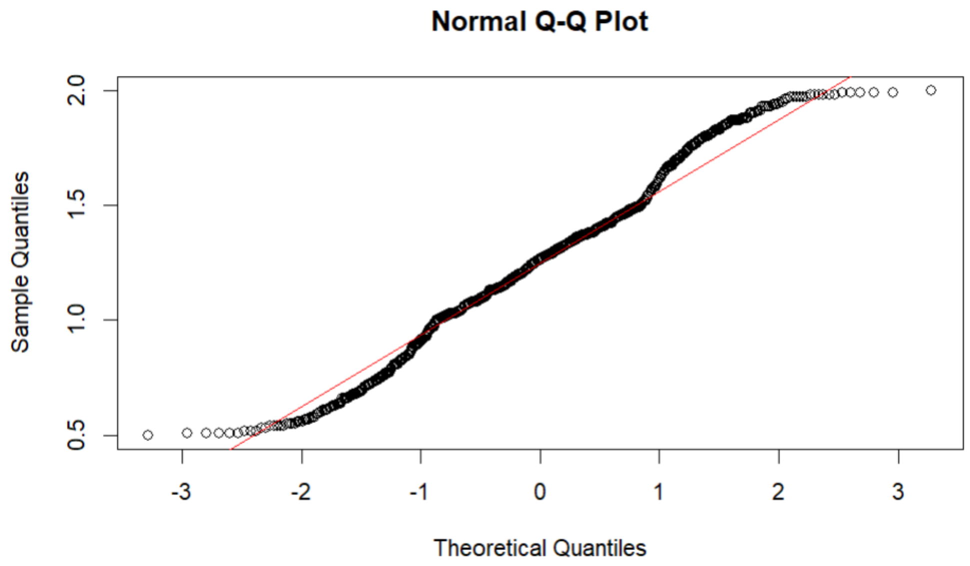
```{r}
# Histogram of Session Duration
hist(data$session_duration,
     main = "Workout Duration Distribution",
     xlab = "Hours per Session",
     col = "lightgreen",
     breaks = 20)
```



The distribution of workout duration is shown using a histogram. Most of the members workout duration is fall between 1.0 to 1.5 hours. It seems to mean that the workout duration is normally distributed, we plot a normal QQ plot and do a Shapiro test to get a further understanding.

````
```{r}
shapiro.test(data$session_duration)
qqnorm(data$session_duration)
qqline(data$session_duration, col = "red")
```
````

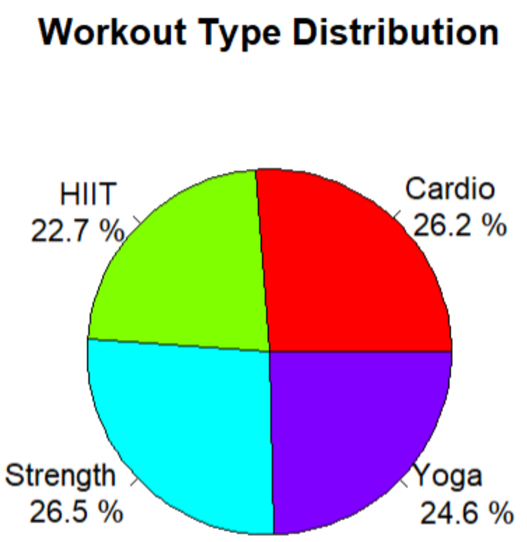
```
        Shapiro-Wilk normality test

data:  data$session_duration
W = 0.98563, p-value = 3.509e-08
```



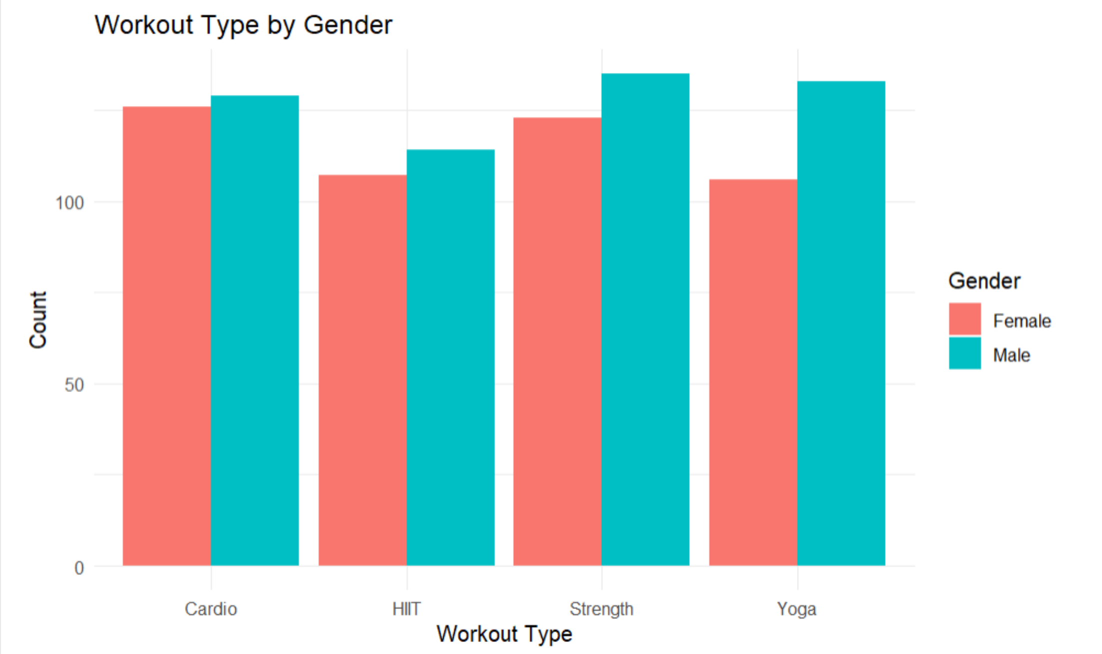
From Shapiro-Wilk test, the p-values is greater than 0.05, indicating that workout duration is normally distributed. Normally QQ plot further support this conclusion.

````
```{r}
#Pie Chart of Workout Type
workout_table <- table(data$workout_type)
workout_table
pie(workout_table,
    main = "Workout Type Distribution",
    col = rainbow(length(workout_table)),
    labels = paste(names(workout_table), "\n", round(100 * workout_table/sum(workout_table), 1), "%"))
```
````

```
 Cardio     HIIT Strength     Yoga
    255      221      258      239
```



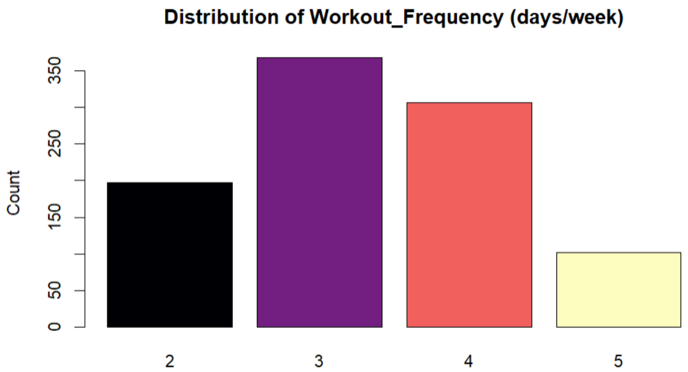

```{r}
#Gender and exercise type
library(ggplot2)

ggplot(data, aes(x = workout_type, fill = Gender)) +
  geom_bar(position = "dodge") +
  labs(title = "Workout Type by Gender", x = "Workout Type", y = "Count") +
  theme_minimal()

```

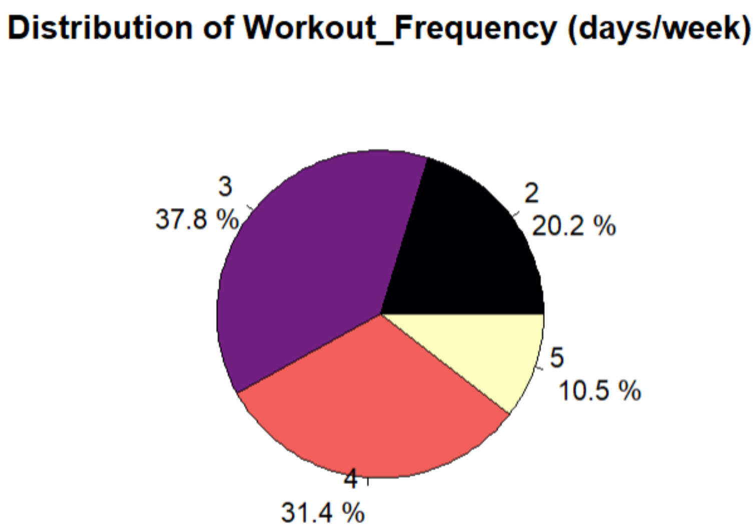


The distribution of workout type is quite evenly, almost each has been chosen by 25% of the members whatever it is grouped by gender or not.

```{r}
#Distribution of Workout_Frequency (days/week)
workout_frequency_table <- table(data$workout_frequency_days_week)
workout_frequency_table

barplot(workout_frequency_table,
        main = "Distribution of Workout_Frequency (days/week)",
        col = magma(length(workout_frequency_table)),
        ylab = "Count")


pie(workout_frequency_table,
    main = "Distribution of Workout_Frequency (days/week)",
    col = magma(length(workout_frequency_table)),
    labels = paste(names(workout_frequency_table), "\n", round(100 *
workout_frequency_table/sum(workout_frequency_table), 1), "%"))

```

```
  2   3   4   5 
197 368 306 102 
```



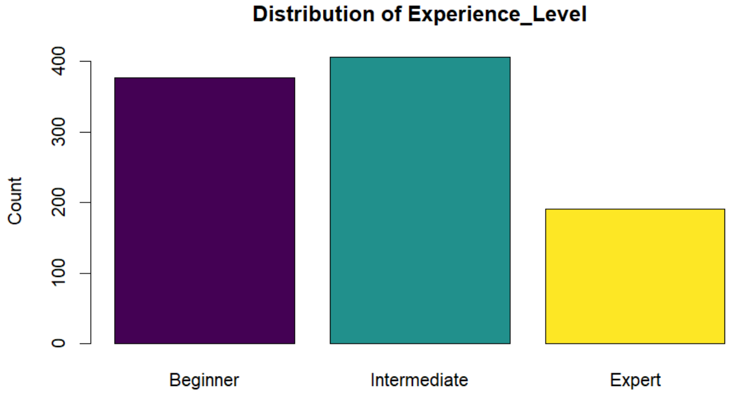


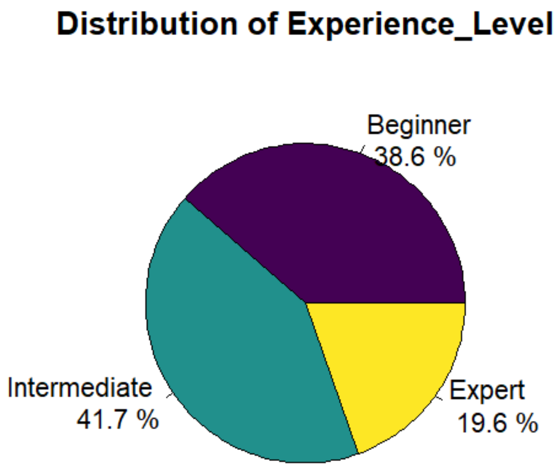
The bar plot and pie chart shows that most of the members choose to work out 3 or 4 days per week. Only a very small number of members (10.5%) choose to work out 5 days per week

```{r}
#Distribution of Experience_Level
experience_level_table <- table(data$experience_level )
experience_level_table

barplot(experience_level_table,
        main = "Distribution of Experience_Level",
        col = viridis(length(experience_level_table)),
        ylab = "Count")


pie(experience_level_table,
    main = "Distribution of Experience_Level",
    col = viridis(length(experience_level_table)),
    labels = paste(names(experience_level_table), "\n", round(100 *
experience_level_table/sum(experience_level_table), 1), "%"))
```






```
Beginner Intermediate       Expert
     376          406          191
```

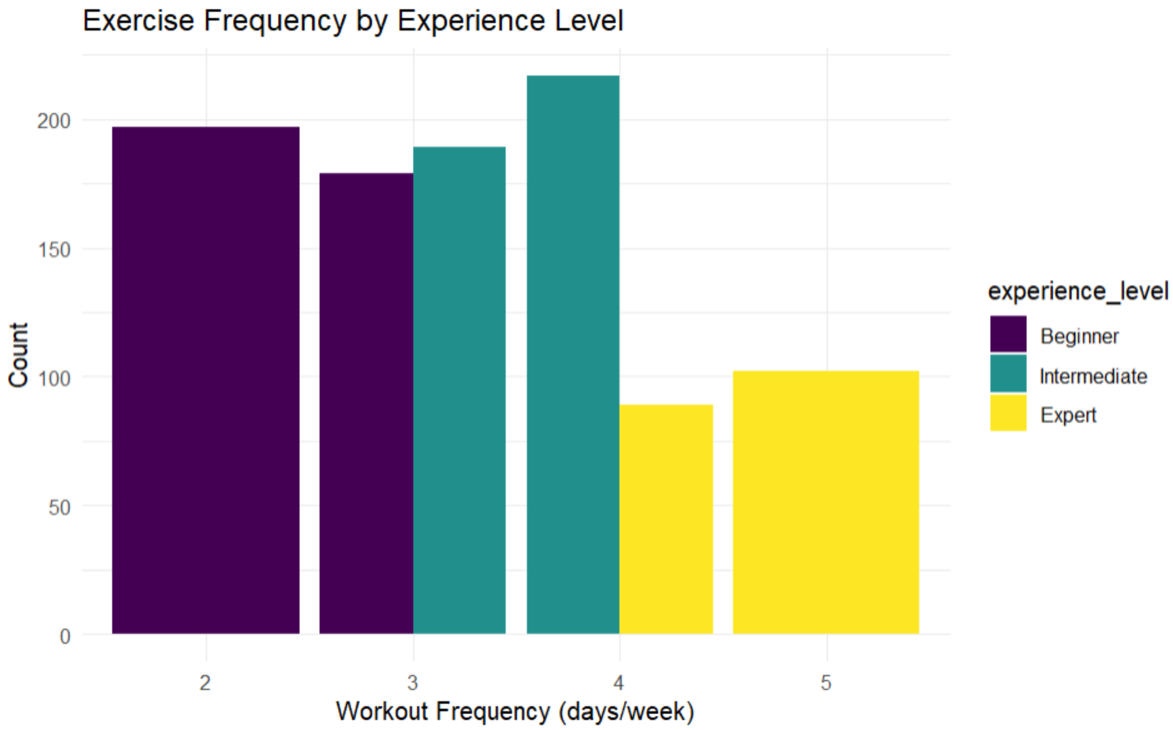
The bar plot and pie chart shows that most of the members are beginner and intermediate trainer, only 19.6% of members are expert trainer.

It seems that there is a relationship between experience level and work out frequency, we try to plot a bar chart of distribution of workout frequency grouped by experience level:

```
```{r}
library(ggplot2)

ggplot(data, aes(x = as.factor(workout_frequency_days_week), fill = experience_level)) +
  geom_bar(position = "dodge") +
  labs(title = "Exercise Frequency by Experience Level",
       x = "Workout Frequency (days/week)",
       y = "Count") +
  theme_minimal() +
  scale_fill_viridis_d()


```
```



The bar chart shows that beginner members always choose to workout 2 or 3 days per week, while intermediate members always choose to workout 3 or 4 days per week, expert members always workout 4 or 5 days per week.

2.2 Descriptive Statistics

Skewness explanation：

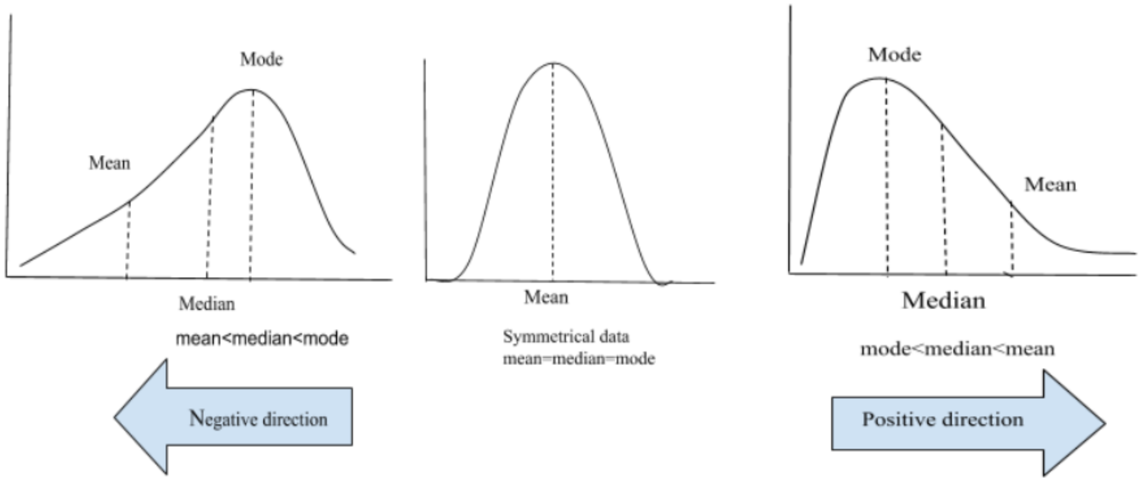
Skewness measures the symmetry of the data distribution:

Skewness > 0 (positive skewness): the data is right-skewed, with most values concentrated on the left and a long tail on the right.

Skewness < 0 (negative skewness): the data is left-skewed, with most values concentrated on the right and a long tail on the left.

Skewness ≈ 0: the data is approximately symmetrical, but not necessarily a perfect normal distribution.

(Dodos & Dodos, 2021)



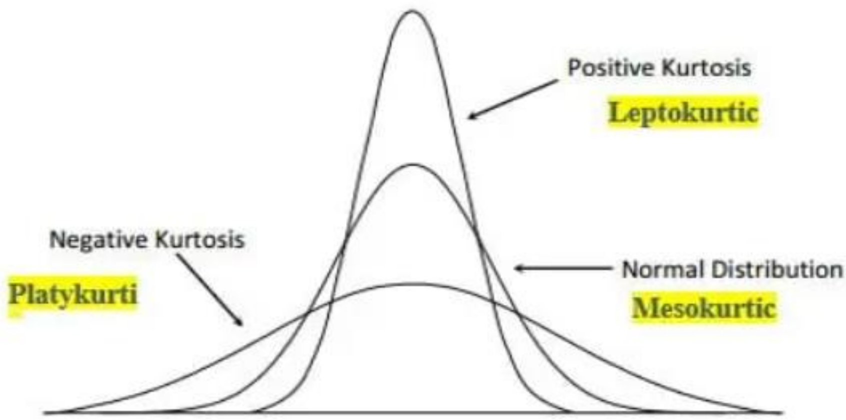
Excess Kurtosis explanation：

Kurtosis measures the thickness of the tail of the data distribution:

Kurtosis > 0 (leptokurtic): The tail is thicker than the normal distribution, with more extreme values (peaked).

Kurtosis < 0 (platykurtic): The tail is thinner than the normal distribution, with fewer extreme values (flat-topped).

Kurtosis ≈ 0: Close to the normal distribution. (Prateek, 2022)



Kurtosis Types

```{r}
#Central Tendency
central_vars <- c("age", "height_m", "weight_kg", "session_duration_hours", "calories_burned", "fat_percentage", "water_intake_liters")

for (v in central_vars) {
  cat("\nVariable:", v, "\n")
  cat("Mean:", mean(data[[v]]), "\n")
  cat("Median:", median(data[[v]]), "\n")
  cat("Mode:", names(which.max(table(data[[v]]))), "\n")
  cat("Standard Deviation:", sd(data[[v]]), "\n")
  cat("Variance:", var(data[[v]]), "\n")
  cat("Skewness:", skewness(data[[v]]), "\n")
  cat("Kurtosis:", kurtosis(data[[v]]), "\n")
}
```

```
Variable: age
Mean: 38.68345
Median: 40
Mode: 43
Standard Deviation: 12.18093
Variance: 148.375
Skewness: -0.07762405
Kurtosis: -1.218668

Variable: height_m
Mean: 1.72258
Median: 1.71
Mode: 1.62
Standard Deviation: 0.1277199
Variance: 0.01631237
Skewness: 0.3378143
Kurtosis: -0.7307513

Variable: weight_kg
Mean: 73.85468
Median: 70
Mode: 57.7
Standard Deviation: 21.2075
Variance: 449.7581
Skewness: 0.7700042
Kurtosis: -0.03610784

Variable: session_duration_hours
Mean: 1.256423
Median: 1.26
Mode: 1.03
Standard Deviation: 0.3430335
Variance: 0.117672
Skewness: 0.02568165
Kurtosis: -0.3605989
```

```
Variable: calories_burned
Mean: 905.4224
Median: 893
Mode: 883
Standard Deviation: 272.6415
Variance: 74333.4
Skewness: 0.2774636
Kurtosis: -0.06795862

Variable: fat_percentage
Mean: 24.97677
Median: 26.2
Mode: 28.1
Standard Deviation: 6.259419
Variance: 39.18032
Skewness: -0.6332675
Kurtosis: -0.3488977

Variable: water_intake_liters
Mean: 2.626619
Median: 2.6
Mode: 3.5
Standard Deviation: 0.6001719
Variance: 0.3602064
Skewness: 0.07125966
Kurtosis: -1.025288
```

Some descriptive statistic has been performed to numerical column to catch some pattern that are hard to discover from graph (data visualization). Mean, median, mode, standard deviation, variance, skewness, and kurtosis is found to understand the data.

1. Age

The average age of gym members in the data set is approximately 38.68 years, with a median of 40 and a mode of 43. These three measures of central tendency are relatively close to each other, suggesting a fairly balanced age distribution. The standard deviation of 12.18 years indicates a moderately wide range of ages among members, reflecting a diverse user base in terms of age. The skewness value of -0.078 implies that the distribution is nearly symmetric, with a very slight negative skew, meaning there are marginally more older individuals than younger ones, but the effect is negligible. The kurtosis value of -1.22 indicates a relatively flat distribution compared to a normal distribution, suggesting fewer individuals at the extreme age values. Overall, age is evenly spread out without any significant concentration at either end, and the distribution appears approximately normal with slight platykurtic behavior.

2. Height

The mean height of gym members is 1.72 meters, while the median is slightly lower at 1.71 meters, and the most frequently occurring height is 1.62 meters. The standard deviation is around 0.13 meters, implying that most members have similar heights, with only minor variation. The positive skewness of 0.34 suggests a slightly right skewed, meaning there are a few members who are significantly taller than the average, which slightly raises the mean. The kurtosis of -0.73 points to a relatively flat distribution, where extreme height values are less common than in a normal distribution. This indicates that although there are a few tall individuals, most members fall within a relatively narrow range of average heights, and the overall height distribution is fairly stable and predictable.

3. Weight

The average weight of members is 73.85 kilograms, with a median of 70 and a mode of 57.7. The relatively high standard deviation of 21.21 kg and variance of 449.76 reflect a large spread of values, indicating considerable diversity in body weights across the gym population. The skewness value of 0.77 reveals a noticeable positive skew, meaning there are several individuals with significantly higher weights pulling the mean upward, even though the median remains relatively moderate. This suggests that while many members have average or below-average weights, a subset of heavier individuals is creating a longer right tail in the distribution. The kurtosis is nearly zero (-0.036), meaning the distribution's tail behavior is similar to that of a normal distribution. This implies a typical concentration of data points near the mean, with the presence of outliers not being excessively extreme.

4. Session Duration (hours)

The mean and median session duration are nearly identical at 1.26 hours, while the mode is slightly lower at 1.03 hours. This tight alignment among mean, median, and mode indicates a highly symmetric distribution. The standard deviation is relatively small (0.34 hours), demonstrating that most members engage in sessions of similar duration, with little variability. The very low skewness value (0.026) confirms the symmetry, and the slightly negative kurtosis (-0.36) indicates a slightly flatter distribution compared to the normal curve. Overall, session duration is consistent across the gym population, showing that members tend to have standardized workout times, possibly influenced by gym schedules, time availability, or recommended training plans.

5. Calories Burned

The mean number of calories burned per session is approximately 905.42, with a median of 893 and a mode of 883. These closely aligned values suggest a symmetric distribution with a central concentration of values. The standard deviation of 272.64 kcal is relatively large, indicating a notable range of calorie expenditure depending on workout type, intensity, and duration. The skewness is mildly positive (0.28), suggesting a slight right tail where some users burn significantly more calories, perhaps due to longer or more intense sessions. The kurtosis is close to zero (-0.068), indicating a normal distribution of values in terms of outlier frequency. This variable is particularly informative when evaluating workout intensity and energy expenditure across the gym population, and it could be a valuable predictor in regression analyses involving physical progress or fitness outcomes.

6. Fat Percentage

The average fat percentage is around 24.98%, with a median of 26.2% and a mode of 28.1%, indicating a slightly left-skewed distribution. The skewness value of -0.63 confirms this, showing a tendency for more individuals to have higher fat percentages than lower ones, but with a long left tail where some users have very low fat levels. The standard deviation is 6.26%, and the variance is 39.18, both suggesting a moderate degree of variation among users. The slightly negative kurtosis (-0.35) suggests a relatively flat distribution, with fewer individuals at the extremes compared to a normal distribution. These values point to a diverse range of body compositions among members, which may reflect varying fitness goals, experience levels, and genetic differences. This variable is often considered a key health metric and can be important in predicting outcomes such as weight loss, performance improvement, or cardiovascular health.

7. Water Intake (liters)

The mean daily water intake is approximately 2.63 liters, with a median of 2.6 liters and a mode of 3.5 liters. These closely aligned central values suggest a fairly symmetric distribution. The standard deviation is 0.60 liters, indicating that most members have similar water consumption habits. The skewness value of 0.071 is almost negligible, further confirming the symmetry of the distribution. However, the kurtosis value of -1.03 is quite low, suggesting a flatter-than-normal distribution, where extreme values are less common and data points are more evenly spread. This may imply that although users follow relatively consistent hydration practices, there is no strong central tendency or dominant behavior. Water intake is a useful variable in assessing users' hydration habits, which may influence energy levels, performance, and general health outcomes.x

```{r}
# Z-score for Calories_Burned
data$z_calories <- scale(data$calories_burned)
head(data$z_calories)

# z > 2 or z < -2 often considered an outlier

```

```
            [,1]
[1,]  1.49492124
[2,] -0.08224134
[3,] -0.83781226
[4,] -1.36964616
[5,] -1.28161848
[6,]  0.77236071
```

Standardize calories_burned by using z-score standardization method. Z > 2 or Z < -2 often considered as an outlier.

```{r}
library(corrplot)

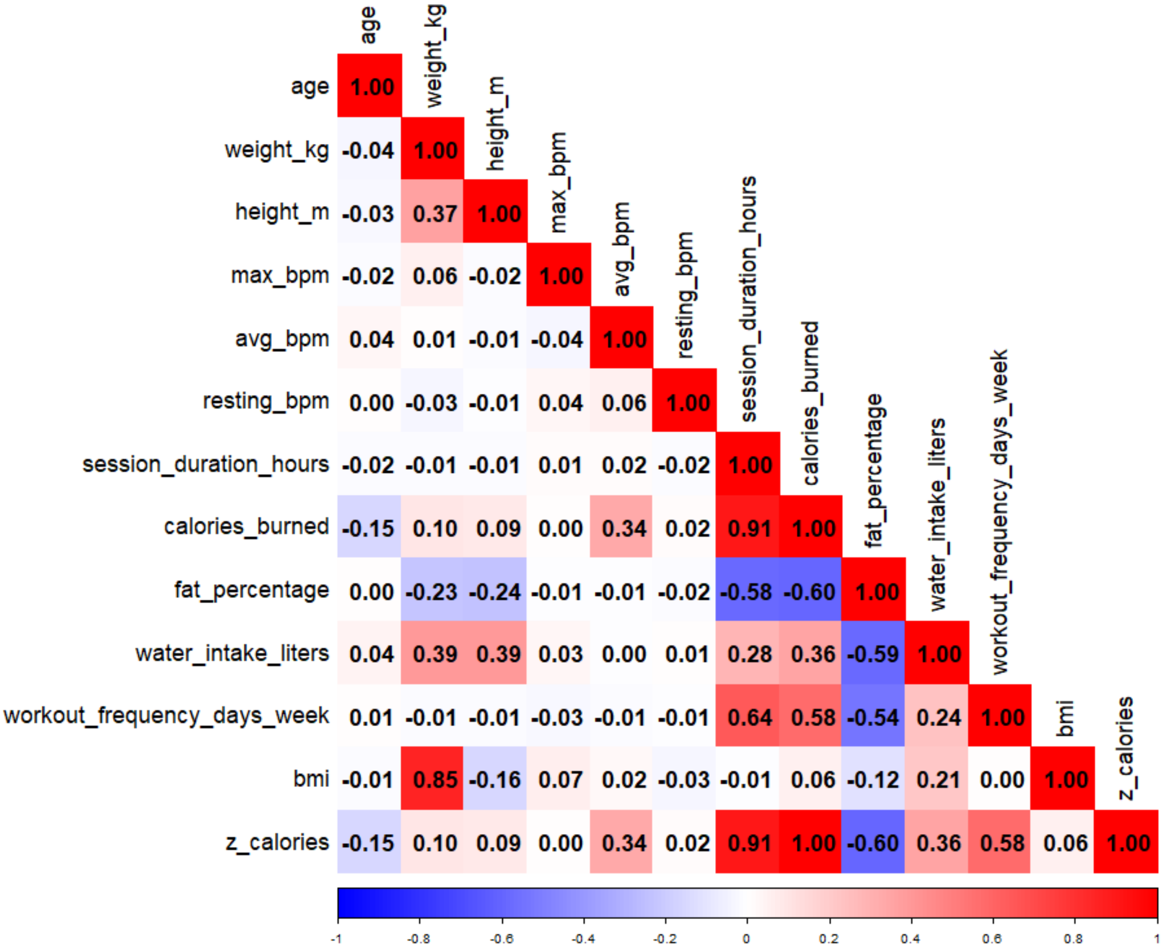
num_vars <- data[, sapply(data, is.numeric)]

cor_matrix_pearson <- cor(num_vars, use = "complete.obs", method = "pearson")
png('C:/University/Year 3 Sem 1/Statistical Programming using R/Assignment/pearson_corrplot.png',
    width = 1000, height = 1000)
corrplot(cor_matrix_pearson, method = "color", type = "lower",
         tl.col = "black", tl.cex = 1.5, addCoef.col = "black",
         number.cex = 1.5, col = colorRampPalette(c("blue", "white", "red"))(200))
dev.off()

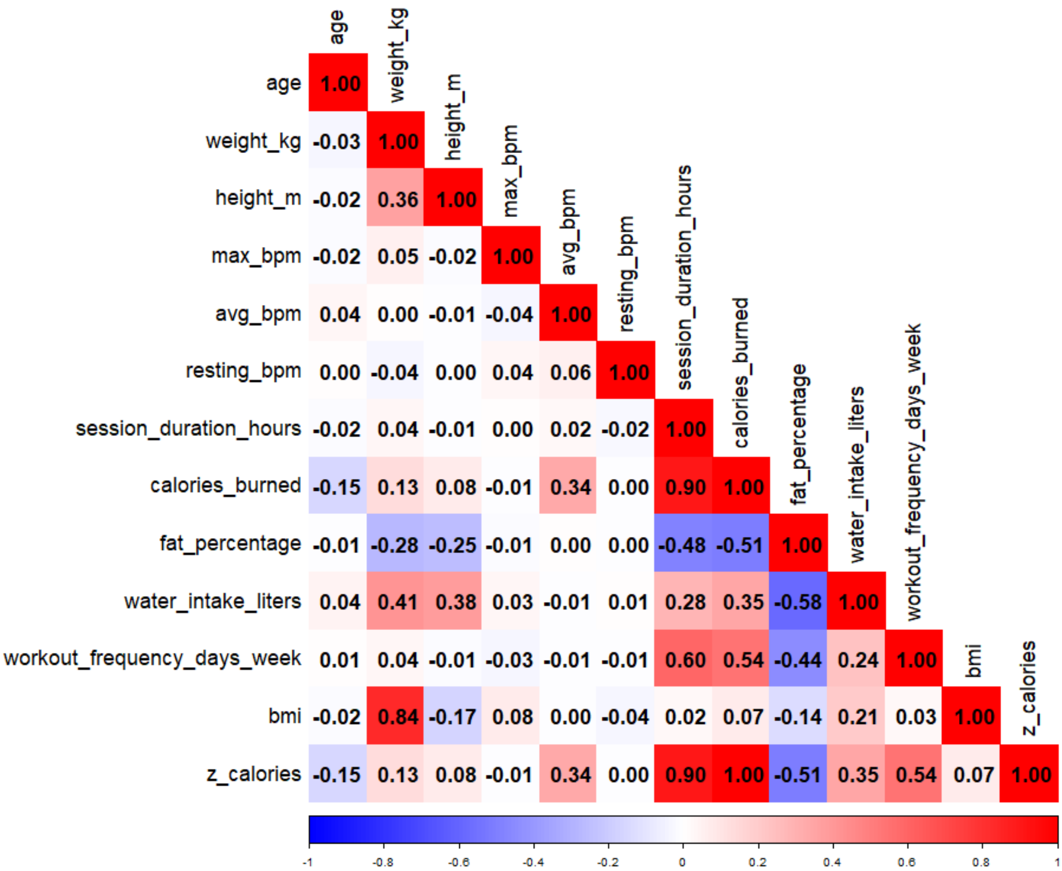
cor_matrix_spearman <- cor(num_vars, use = "complete.obs", method = "spearman")
png('C:/University/Year 3 Sem 1/Statistical Programming using R/Assignment/spearman_corrplot.png',
    width = 1000, height = 1000)
corrplot(cor_matrix_spearman, method = "color", type = "lower",
         tl.col = "black", tl.cex = 1.5, addCoef.col = "black",
         number.cex = 1.5, col = colorRampPalette(c("blue", "white", "red"))(200))
dev.off()

```

Pearson correlation matrices:



Spearman correlation matrices:



I computed and visualized both the **Pearson** and **Spearman** correlation matrices to examine the relationships among numeric variables. The darker the color, the more relevant it is. The Pearson correlation, which captures **linear relationships**, revealed strong positive associations between variables such as weight_kg and bmi (0.85), calories_burned and session_duration_hours (0.91). It indicates that the heavier the member, the higher the BMI value. The longer the fitness session duration hour, the more calories consumed. There are moderate positive associations between variables such as session_duration_hours and workout_frequency_days_week (0.64), calories_burned and workout_frequency_days_week(0.58).

To validate whether any **non-linear but monotonic** relationships existed, I also computed the Spearman correlation. The overall structure was similar to Pearson's, indicating that most variable relationships in this dataset are approximately linear. Therefore, Pearson correlation results were used in the main analysis and interpretation.

3.0 Problem statement

This study will address the following questions:

1. Estimating Member Calorie Burned (Relating Discrete Probability to Normal Probability)

2. Can Small Samples Estimate true population mean of Calorie Burn? (The Sampling Distribution of Means)

3. Does average exercise heart rate (avg_bpm) exceed the recommended value of 120? (One-Sample t-Test)

4. Is there a significant difference in the heart rate of gym members before and after exercise? (Paired-Samples t-Test)

5. Is the average heart rate different in people with high BMI vs normal BMI? (Two-Sample t-Tests (Mann-Whitney U Test))

6. Is there a significant difference in average heart rate between men and women? (Two-Sample t-Tests (Mann-Whitney U Test))

7. Do people who exercise regularly burn more calories than those who do not exercise regularly? (Two-Sample t-Tests (Mann-Whitney U Test))

8. What is the Confidence Intervals for Population Means of fat percentage?

9. What is Confidence Intervals for Population Proportions of members with healthy BMI?

10. Examine whether there is a significant difference in the BMI by different workout frequency (2, 3, 4, 5 days / week). (One -Way ANOVA)

11. Examine whether there is a significant difference in the average calories burned by different types of exercise (Yoga, HIIT, Cardio, Strength). (One -Way ANOVA)

12. Examine whether there is a significant difference in the average calories burned by different types of exercise (Yoga, HIIT, Cardio, Strength) and gender. (Two -Way ANOVA)

13. How different factors affect calories burned? (Multiple Linear Regression Model)

3.1 Estimating Member Calorie Burned (Relating Discrete Probability to Normal Probability)

Gym management wants to know how common it is for members to do high-calorie burn training. Specifically, they want to know: If we randomly sample 100 gym members, what is the probability that at least 60 of them will burn more than 800 calories in a single workout? We want to run an advertisement that shows that our gym program can significantly help members burn more calories.

````
```{r}
#Problem Statement Start:
#Q1
#add a column
data$burns_800plus <- ifelse(data$calories_burned >= 800, 1, 0)
p_hat <- mean(data$burns_800plus)
p_hat

n <- 100
# binomial compute P(X ≥ 60)
d_binom <- sum(dbinom(60:100, size = n, prob = p_hat))
cat("P(X ≥ 60) using Binomial:", d_binom , "\n")

mu <- n * p_hat
sigma <- sqrt(n * p_hat * (1 - p_hat))

# Normal approximation P(X ≥ 60)
p_normal <- pnorm(59.5, mean = mu, sd = sigma, lower.tail = FALSE)
cat("P(X ≥ 60) using Normal Approximation:", p_normal , "\n")


```
````

```
[1] 0.647482
P(X ≥ 60) using Binomial: 0.8636349
P(X ≥ 60) using Normal Approximation: 0.8640102
```

To compute the probability that **at least 60 out of 100 randomly selected gym members** burn **800 or more calories** in a session, first created a binary variable burns_800plus indicating whether each member's calories burned was at least 800 (1) or not (0). The sample proportion (p_hat) of such high-calorie-burning members was approximately **0.6475**.

Then calculated the probability of $X \geq 60$ under two approaches:

1. **Binomial Distribution**:

Assuming $X \sim \text{Binomial}(n = 100, p = 0.6475)$, we computed $P(X \geq 60)$, as the sum of binomial probabilities from 60 to 100. The result was approximately **0.8636**.

2. **Normal Approximation to Binomial**:

Using the Central Limit Theorem, the distribution of X was approximated by a normal distribution with $\mu = np, \sigma = \sqrt{np(1-p)}$. Applying a continuity correction, we estimated $P(X \geq 60) \approx P(X > 59.5)$. The normal approximation yielded a probability of approximately **0.8640**, which is very close to the exact binomial result.

Conclusion: The normal approximation provides an accurate estimate in this case due to the large sample size and the value of np and n(1−p) both being sufficiently large. Thus, we conclude that the probability that at least 60 out of 100 members burn over 800 calories is about 86%. Marketing can highlight that a significant portion of members achieve substantial calorie burn, enhancing gym value.

3.2 Can Small Samples Estimate true population mean of Calorie Burn? (The Sampling Distribution of Means)

The database system of the gym is broken, and I lost all the data and can only take 30 samples from the members at a time. Can the sample mean calorie burn value stably estimate the true average of the whole population?

````
```{r}
#Q2
population <- data$calories_burned
population_mean <- mean(population)
print(paste("Population Mean:", population_mean))

# Plot histogram of the population
hist(population, breaks = 30, col = "lightblue", border = "black",
     main = "Histogram of Population Distribution",
     xlab = "Value", ylab = "Frequency")

set.seed(123)
sample_means <- replicate(1000, {
  sample_30 <- sample(population, 30, replace = TRUE)
  mean(sample_30)
})

hist(sample_means, breaks = 30, col = "skyblue", border = "white",
     freq = FALSE,
     main = "Sampling Distribution of the Mean",
     xlab = "Sample Mean", ylab = "Density")

curve(dnorm(x, mean = mean(sample_means), sd = sd(sample_means)),
      col = "red", lwd = 2, add = TRUE)

sample_mean <- mean(sample_means)
print(paste("Estimate Population Means:", sample_mean))


shapiro.test(population)
shapiro.test(sample_means)

```
````

```
[1] "Population Mean: 905.422404933196"
[1] "Estimate Population Means: 904.536266666667"

        Shapiro-Wilk normality test

data:  population
W = 0.99176, p-value = 2.982e-05


        Shapiro-Wilk normality test

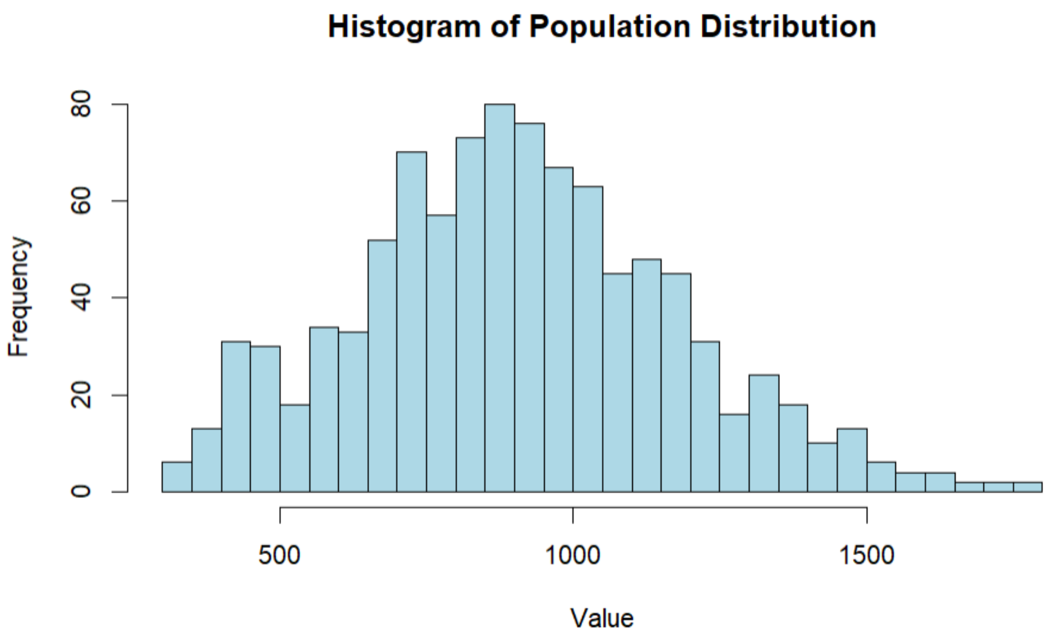
data:  sample_means
W = 0.99863, p-value = 0.6404
```

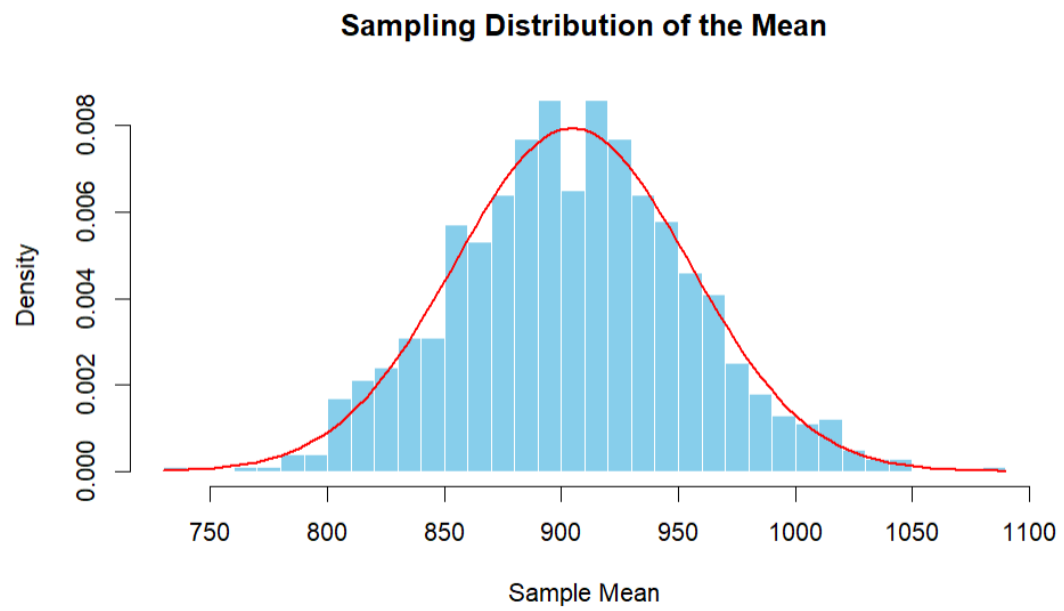
In this analysis, we explored the sampling distribution of the average calories burned per session. The population consisted of all members' calories_burned values, with a true population mean of approximately 905.42.

We simulated the process of random sampling by:

1. Repeatedly drawing 1000 samples, each of 30 members, with replacement.

2. Calculating the mean of each sample to form the sampling distribution of the sample mean.





Although the original population is not normally distributed (we can verify this by Shapiro test, the p-value is 2.982×10^{-5}, which is lower than 0.05 indicating not normally distributed), the distribution of the sample means is approximately normal (p-value of Shapiro test is 0.6404, which is greater than 0.05, indicating normally distributed). This supports the **Central Limit Theorem**, which states that the sampling distribution of the mean tends toward normality as the sample size increases (in this case, n = 30 is already sufficient).

The sample-based estimate of the population mean was approximately **904.54**, which is very close to the true population mean (905.42), confirming the reliability of the sampling approach. This means stakeholders can make confident decisions using small sample surveys, saving time and resources while maintaining accuracy.

3.3 Does average exercise heart rate (avg_bpm) exceed the recommended value of 120? (One-Sample t-Test)

One-sample t-tests need to meet the normal distribution assumption. However, since the sample size is large (n = 974), the Central Limit Theorem (CLT) applies. This theorem states that the sampling distribution of the mean approaches normality as the sample size increases, regardless of the population distribution. Therefore, normality assumptions can be reasonably relaxed in this case, and the t-tests remain valid and reliable.

Statistical method: One-Sample t-Test

H0: $\mu 1 = 120$ (the average exercise heart rate is equal to 120)

H1: $\mu 1 > 120$ (the average exercise heart rate exceed the recommended value of 120)

````
```{r}
#Q3
threshold <- 120

# One-Sample t-Test
t_test <- t.test(data$avg_bpm , mu = threshold, alternative = "greater")
t_test

```
````

```
        One Sample t-test

data:  data$avg_bpm
t = 51.68, df = 972, p-value < 2.2e-16
alternative hypothesis: true mean is greater than 120
95 percent confidence interval:
 143.0095      Inf
sample estimates:
mean of x
 143.7667
```

Since the p-value is very small, we **reject the null hypothesis**, indicating that the average exercise heart rate significantly exceed the recommended value of 120, means that most gym members reach the recommended exercise heart rate of 120.

3.4 Is there a significant difference in the heart rate of gym members before and after exercise? (Paired-Samples t-Test)

Paired-sample t-test:

Paired-sample t-tests need to meet the normal distribution assumption. However, since the sample size is large (n = 974), the Central Limit Theorem (CLT) applies. This theorem states that the sampling distribution of the mean approaches normality as the sample size increases, regardless of the population distribution. Therefore, normality assumptions can be reasonably relaxed in this case, and the t-tests remain valid and reliable.

H0: $\mu d = 0$ (The mean difference in the heart rate before and after exercise is zero)

H1: $\mu d \neq 0$ (The mean difference in the heart rate before and after exercise is not zero)

```
```{r}
# Paired t-test
paired_test <- t.test(data$avg_bpm, data$resting_bpm, paired = TRUE)
paired_test

```
```

```
        Paired t-test

data:  data$avg_bpm and data$resting_bpm
t = 161.87, df = 972, p-value < 2.2e-16
alternative hypothesis: true mean difference is not equal to 0
95 percent confidence interval:
 80.55508 82.53228
sample estimates:
mean difference
       81.54368
```

Since p-value is very small, we **reject null hypothesis**, indicating that the mean difference in the heart rate before and after exercise is not zero. 95% confidence interval of the mean difference is between 80.56 and 82.53 beats per minute. The average heart rate of fitness members during exercise is significantly higher than the resting heart rate (mean difference is 81.54 beats per minute), which is in line with physiological expectations, indicating that exercise significantly increases heart rate levels.

3.5 Is the average heart rate different in people with high BMI vs normal BMI? (Two-Sample t-Tests (Mann-Whitney U Test))

Statistical method: Two-Sample t-Test

H0: $\mu 1 = \mu 2$ (There is no difference in average heart rate between normal BMI members and high BMI members)

H1: $\mu1 \neq \mu2$ (There is significant difference in average heart rate between normal BMI members and high BMI members)

```
```{r}
#Q5 (two-sample t test)
normal_BMI <- subset(data, bmi >= 18.5 & bmi <= 24.9)$avg_bpm
high_BMI <- subset(data, bmi > 25)$avg_bpm


shapiro_normal <- shapiro.test(normal_BMI)
shapiro_high <- shapiro.test(high_BMI)
shapiro_normal
shapiro_high

if (shapiro_normal$p.value > 0.05 & shapiro_high$p.value > 0.05) {
  print("Both groups are normally distributed")
} else {
  print("Either one or both groups are NOT normally distributed")
}

```
```

```
        Shapiro-Wilk normality test

data:  normal_BMI
W = 0.95845, p-value = 1.159e-08


        Shapiro-Wilk normality test

data:  high_BMI
W = 0.9493, p-value = 4.696e-11

[1] "Either one or both groups are NOT normally distributed"
```

Since the Shapiro-Wilk test fails (the p-value is much less than 0.05), it means that the sample does not follow a normal distribution, so the two sample t-test (regardless of whether the variance is equal) is not applicable. Therefore, a non-parametric Mann-Whitney U Test was conducted：

Although the test is referred to as the Wilcoxon Rank-Sum Test in R, it is functionally equivalent to the Mann-Whitney U Test when used to compare two independent samples. In this analysis, since each group contains more than 30 observations, the Mann-Whitney U Test is more appropriate terminology.

```
```{r}
wilcox_test_result <- wilcox.test(normal_BMI, high_BMI)

print(wilcox_test_result)

if (wilcox_test_result$p.value < 0.05) {
  decision <- "Reject the null hypothesis (H0)"
} else {
  decision <- "Fail to reject the null hypothesis (H0)"
}
cat("Decision: ", decision, "\n")
```
```

```
        Wilcoxon rank sum test with continuity correction

data:  normal_BMI and high_BMI
W = 78528, p-value = 0.7412
alternative hypothesis: true location shift is not equal to 0

Decision:  Fail to reject the null hypothesis (H0)
```

Since p-value is 0.7412, larger than 0.05, we **failed to reject null hypothesis,** indicating that there is no significant difference in average heart rate between normal BMI members and high BMI members.

3.6 Is there a significant difference in average heart rate between men and women? (Two-Sample t-Tests (Mann-Whitney U Test))

Statistical method: Two-Sample t-Test

H0: $\mu1 = \mu2$ (There is no difference in average heart rate between men and women)

H1: $\mu1 \neq \mu2$ (There is significant difference in average heart rate between men and women)

```
group_male <- data$avg_bpm[data$gender == "Male"]
group_female <- data$avg_bpm[data$gender == "Female"]

shapiro_male <- shapiro.test(group_male)
shapiro_female <- shapiro.test(group_female)
shapiro_male
shapiro_female

if(shapiro_male$p.value > 0.05 & shapiro_female$p.value > 0.05){
  print("Both groups are normally distributed")
}else{
  print("Either one or both groups are NOT normally distributed")
}
```

```
        Shapiro-Wilk normality test

data:  group_male
W = 0.95224, p-value = 8.57e-12


        Shapiro-Wilk normality test

data:  group_female
W = 0.95256, p-value = 5.076e-11

[1] "Either one or both groups are NOT normally distributed"
```

Since the Shapiro-Wilk test fails (the p-value is much less than 0.05), it means that the sample does not follow a normal distribution, so the two sample t-test (regardless of whether the variance is equal) is not applicable. Therefore, a non-parametric Mann-Whitney U Test was conducted：

```
```{r}
wilcox_test_result <- wilcox.test(group_male, group_female)

print(wilcox_test_result)

if (wilcox_test_result$p.value < 0.05) {
  decision <- "Reject the null hypothesis (H0): Significant difference exists in average heart rate."
} else {
  decision <- "Fail to reject the null hypothesis (H0): No significant difference in average heart rate."
}
cat("Decision: ", decision, "\n")

```
```

```
        Wilcoxon rank sum test with continuity correction

data:  group_male and group_female
W = 119402, p-value = 0.756
alternative hypothesis: true location shift is not equal to 0

Decision:  Fail to reject the null hypothesis (H0): No significant difference in average heart rate.
```

Since p-value is 0.756, larger than 0.05, we **failed to reject null hypothesis,** indicating that there is no significance difference in average heart rate between men and women. Gender-neutral heart rate-based training appears sufficient to maximize our profit.

3.7 Do people who exercise regularly burn more calories than those who do not exercise regularly? (Two-Sample t-Tests (Mann-Whitney U Test))

Statistical method: Two-Sample t-Test

H0: $\mu1 = \mu2$ (There is no difference in calories burn between people who exercise regularly and not regularly)

H1: μ1 ≠ μ2 (There is significant difference in calories burn between people who exercise regularly and not regularly)

```
```{r}
#Q2:  Do people who exercise regularly burn more calories than those who do not exercise regularly?
# add new column
data$regular_exercise <- ifelse(data$workout_frequency_days_week >= 4, "Regular", "Irregular")


group_regular <- data$calories_burned[data$regular_exercise == "Regular"]
group_irregular <- data$calories_burned[data$regular_exercise == "Irregular"]


shapiro_reg <- shapiro.test(group_regular)
shapiro_irreg <- shapiro.test(group_irregular)
shapiro_reg
shapiro_irreg

if (shapiro_reg$p.value > 0.05 & shapiro_irreg$p.value > 0.05) {
  print("Both groups are normally distributed")
} else {
  print("Either one or both groups are NOT normally distributed")
}
```
```

```
	Shapiro-Wilk normality test

data:  group_regular
W = 0.97994, p-value = 1.966e-05


	Shapiro-Wilk normality test

data:  group_irregular
W = 0.98617, p-value = 3.403e-05

[1] "Either one or both groups are NOT normally distributed"
```

Since the Shapiro-Wilk test fails (the p-value is much less than 0.05), it means that the sample does not follow a normal distribution, so the two sample t-test (regardless of whether the variance is equal) is not applicable. Therefore, a non-parametric Mann-Whitney U Test was conducted：

```{r}
# Wilcoxon Rank-Sum Test (Mann-Whitney U Test)
wilcox_test_result <- wilcox.test(group_regular, group_irregular)
print(wilcox_test_result)

if (wilcox_test_result$p.value < 0.05) {
  decision <- "Reject the null hypothesis (H0)"
} else {
  decision <- "Fail to reject the null hypothesis (H0)"
}
cat("Decision: ", decision, "\n")

```

```
        Wilcoxon rank sum test with continuity correction

data:  group_regular and group_irregular
W = 180792, p-value < 2.2e-16
alternative hypothesis: true location shift is not equal to 0

Decision:  Reject the null hypothesis (H0)
```

Since p-value is very small, we **reject null hypothesis,** indicating that there is a statistically significant difference in calories burned between individuals who exercise regularly (4–5 days/week) and those who do not (2–3 days/week). This suggests that people who exercise more frequently tend to burn more calories. This suggests the benefits of promoting frequent attendance through loyalty programs.

3.8 What is the Confidence Intervals for Population Means of fat percentage?

The confidence interval for the population mean is based on the assumption that the sample is drawn from a normally distributed population. However, due to the large sample size (n = 974), the Central Limit Theorem ensures that the sampling distribution of the mean is approximately normal. Therefore, the confidence interval remains valid even if the original data are not perfectly normally distributed.

Studies have shown that a healthy body fat range is **25-31% for women** and **18-24% for men.** (Barnes, 2024).In this analysis,The author build 95% Confidence Intervals for the population mean of fat percentage to identify whether the fat percentage of gym members is within this healthy standard. Since the population standard deviation is unknown and the sample size is finite, the author used the **t-distribution** for the confidence interval estimation.

```{r}
#Q8
#Confidence Intervals for Population mean of fat percentage
# Male
male_data <- subset(data, gender == "Male")
n <- nrow(male_data)
sample_mean <- mean(male_data$fat_percentage)
sample_sd <- sd(male_data$fat_percentage)

alpha <- 0.05
df <- n - 1
t_critical <- qt(1 - alpha/2, df)

lower_bound <- sample_mean - (t_critical * (sample_sd / sqrt(n)))
upper_bound <- sample_mean + (t_critical * (sample_sd / sqrt(n)))
cat("The 95% confidence interval for the population mean of fat percentage of male is between",
    round(lower_bound, 2), "and", round(upper_bound, 2), "%.\n")


# Female
female_data <- subset(data, gender == "Female")
n <- nrow(female_data)
sample_mean <- mean(female_data$fat_percentage)
sample_sd <- sd(female_data$fat_percentage)

alpha <- 0.05
df <- n - 1
t_critical <- qt(1 - alpha/2, df)

lower_bound <- sample_mean - (t_critical * (sample_sd / sqrt(n)))
upper_bound <- sample_mean + (t_critical * (sample_sd / sqrt(n)))
cat("The 95% confidence interval for the population mean of fat percentage of female is between",
    round(lower_bound, 2), "and", round(upper_bound, 2), "%.\n")
```

```
The 95% confidence interval for the population mean of fat percentage of male is between 22.06 and 23.05 %.
The 95% confidence interval for the population mean of fat percentage of female is between 27.13 and 28.18 %.
```

We are 95% confident that the **true population mean** fat percentage lies between 22.06% and 23.05% for males, and between 27.13% and 28.18% for females.They are both within the healthy standard of the fat percentage, indicating that fitness can help maintain a more ideal body fat range. These intervals also reveal a clear gender difference in average fat percentage, with females having a higher estimated mean.

3.9 What is Confidence Intervals for Population Proportions of members with healthy BMI?

Studies have shown that a **healthy BMI range is between 18.5 and 25** (*Body Mass Index in Adults*, 2024). In this analysis,The author build 95% Confidence Intervals for the population proportions of members with healthy BMI. Since the sampling distribution of proportions approximates normality for large n, the author used the **normal distribution** to construct the confidence interval.

```{r}
n*sample_proportion
n*(1 - sample_proportion)
```

```
[1] 370
[1] 603
```

Both np and n(1−p) exceed 10, supporting the use of the normal approximation.

```
```{r}
#Q9
#Confidence Intervals for Population Proportions of members with healthy BMI
n <- nrow(data)
X <- sum(data$bmi >= 18.5 & data$bmi <= 25)
#p_hat
sample_proportion <- X/n

alpha <- 0.05
z_critical <- qnorm(1 - alpha/2)

standard_error <- sqrt((sample_proportion * (1-sample_proportion)) / n)
lower_bound <- sample_proportion - (z_critical * standard_error)
upper_bound <- sample_proportion + (z_critical * standard_error)

cat("The 95% confidence interval for the population proportion of members with healthy BMI is between",
    round(lower_bound, 4) * 100, "and", round(upper_bound, 4) * 100, "%.\n")
```
```

```
The 95% confidence interval for the population proportion of members with healthy BMI is between 34.98 and 41.08 %.
```

We are 95% confident that the **true proportion** of gym members with a healthy BMI lies between **34.98% and 41.08%**. This suggests that roughly one-third to two-fifths of the gym population falls within the standard healthy BMI range.

3.10 Examine whether there is a significant difference in the BMI by different workout frequency (2, 3, 4, 5 days / week). (One -Way ANOVA)

This analysis aimed to investigate whether gym members who work out at different weekly frequencies (2, 3, 4, or 5 days per week) show statistically significant differences in their **Body Mass Index (BMI)**.

H0: $\mu1 = \mu2 = \mu3 = \mu4$ （All group means are equal.）

H1: at least one μj is different. $j = 1,2,3,4$ (At least one group mean is different.)

```
```{r}
#Q10
#One-Way ANOVA
#Examine whether there is a significant difference in the BMI by different workout frequency (2, 3, 4, 5 days / week).

library(car)
data$workout_frequency_days_week <- as.factor(data$workout_frequency_days_week)


#Shapiro-Wilk
shapiro_results <- by(data$bmi, data$workout_frequency_days_week, shapiro.test)
shapiro_results

#Levene's Test
levene_test <- leveneTest(bmi ~ workout_frequency_days_week, data = data)
levene_test
```
```

```
data$workout_frequency_days_week: 2

        Shapiro-Wilk normality test

data:  dd[x, ]
W = 0.96783, p-value = 0.0001744

-------------------------------------------------------------------
data$workout_frequency_days_week: 3

        Shapiro-Wilk normality test

data:  dd[x, ]
W = 0.9578, p-value = 8.691e-09

-------------------------------------------------------------------
data$workout_frequency_days_week: 4

        Shapiro-Wilk normality test

data:  dd[x, ]
W = 0.95575, p-value = 5.401e-08

-------------------------------------------------------------------
data$workout_frequency_days_week: 5

        Shapiro-Wilk normality test

data:  dd[x, ]
W = 0.97216, p-value = 0.02974

Levene's Test for Homogeneity of Variance (center = median)
       Df F value    Pr(>F)
group   3  12.734 3.627e-08 ***
      969
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

Before running a one-way ANOVA, first checked 2 key assumptions:

1. Normality: The Shapiro-Wilk tests showed that BMI is **not normally distributed** across all workout frequency groups (all p-values < 0.05).

2. Homogeneity of Variance: Levene’s Test indicated a significant difference in variances between the groups (unequal variances.) (due to p < 0.05), violating the equal variance assumption.

Two assumption is violated, one-way ANOVA is not appropriate for this scenario. The author proceeded with the **Kruskal-Wallis test** (non-parametric alternative), which is robust to such violations, as an alternative to ANOVA.

```{r}
# Kruskal-Wallis Test
kruskal_result <- kruskal.test(bmi ~ workout_frequency_days_week, data = data)
kruskal_result

```

```
	Kruskal-Wallis rank sum test

data:  bmi by workout_frequency_days_week
Kruskal-Wallis chi-squared = 2.366, df = 3, p-value = 0.5
```

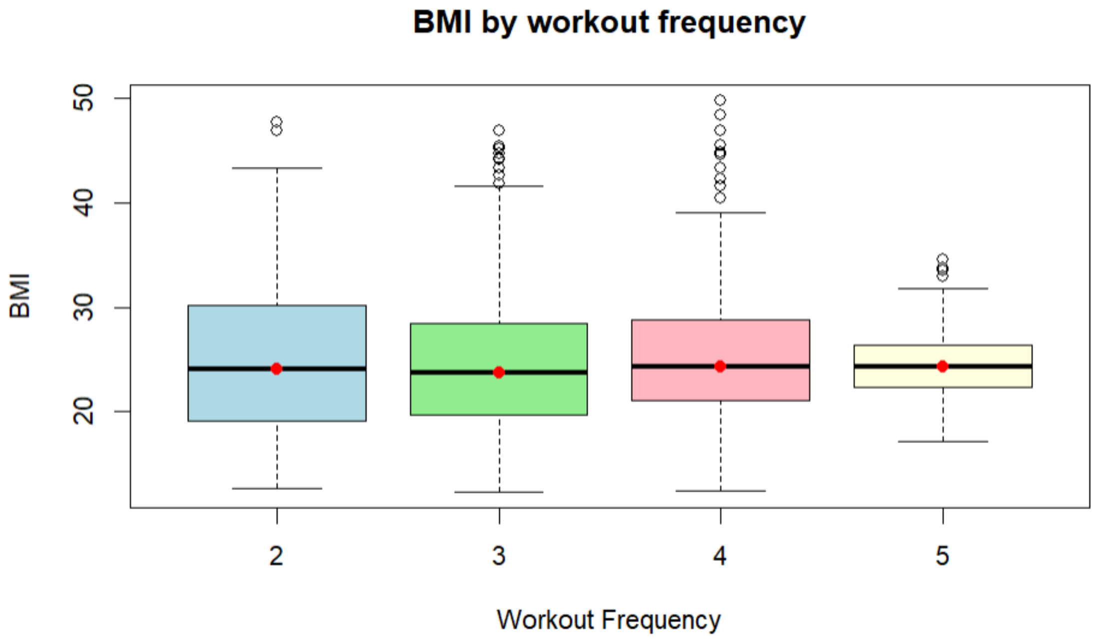
Since p-value is 0.5, larger than 0.05, we **failed to reject null hypothesis, indicating** there is **no statistically significant difference in BMI** across workout frequencies. This suggests that **simply increasing the number of workout days per week (from 2 to 5)** may not lead to a substantial change in BMI on its own. Other factors such as diet, workout intensity, and body composition could be playing more significant roles in determining BMI. This result highlights that **consistency alone isn't the only variable** that influences BMI outcomes. For fitness centers or personal trainers, this underscores the importance of personalized programs that go beyond just frequency recommendations.

The following box-plot visualize the comparison of BMI across workout frequencies:

```{r}
boxplot(bmi ~ workout_frequency_days_week, data = data,
        main = "BMI by workout frequency",
        xlab = "Workout Frequency",
        ylab = "BMI",
        col = c("lightblue", "lightgreen", "lightpink", "lightyellow"))


points(1:length(levels(data$workout_frequency_days_week)),
       tapply(data$bmi, data$workout_frequency_days_week, median),
       col = "red", pch = 19)

```



3.11 Examine whether there is a significant difference in the average calories burned by different types of exercise (Yoga, HIIT, Cardio, Strength). (One -Way ANOVA)

In this analysis, we examined whether there are significant differences in the **average calories burned** across different **types of workouts**: **Yoga**, **HIIT**, **Cardio**, and **Strength training**.

H0: $\mu1 = \mu2 = \mu3 = \mu4$ （All group means are equal.）

H1: at least one μj is different. $j = 1,2,3,4$ (At least one group mean is different.)

```
```{r}
#Q11
#One-Way ANOVA

library(car)
data$workout_type <- as.factor(data$workout_type)


#Shapiro-Wilk
shapiro_results <- by(data$calories_burned, data$workout_type, shapiro.test)
shapiro_results

#Levene's Test
levene_test <- leveneTest(calories_burned ~ workout_type, data = data)
levene_test

```
```

```
data$workout_type: Cardio

        Shapiro-Wilk normality test

data:  dd[x, ]
W = 0.98907, p-value = 0.05108

------------------------------------------------------------
data$workout_type: HIIT

        Shapiro-Wilk normality test

data:  dd[x, ]
W = 0.98673, p-value = 0.03752

------------------------------------------------------------
data$workout_type: Strength

        Shapiro-Wilk normality test

data:  dd[x, ]
W = 0.98022, p-value = 0.00118

------------------------------------------------------------
data$workout_type: Yoga

        Shapiro-Wilk normality test

data:  dd[x, ]
W = 0.98467, p-value = 0.01125

Levene's Test for Homogeneity of Variance (center = median)
       Df F value Pr(>F)
group   3  0.3502  0.789
      969
```

Before running a one-way ANOVA, first checked 2 key assumptions:

1. Normality: The Shapiro-Wilk tests showed **some deviation from normality**, particularly in the HIIT, Strength and Yoga groups (p-values < 0.05), these 3 group are **not normally distributed**.

2. Homogeneity of Variance: Levene's Test yielded **non-significant results** (p = 0.789), suggesting **equal variances** among groups.

Normality assumption is violated, one-way ANOVA is not appropriate for this scenario. The author proceeded with the **Kruskal-Wallis test** (non-parametric alternative), which is robust to such violations, as an alternative to ANOVA.

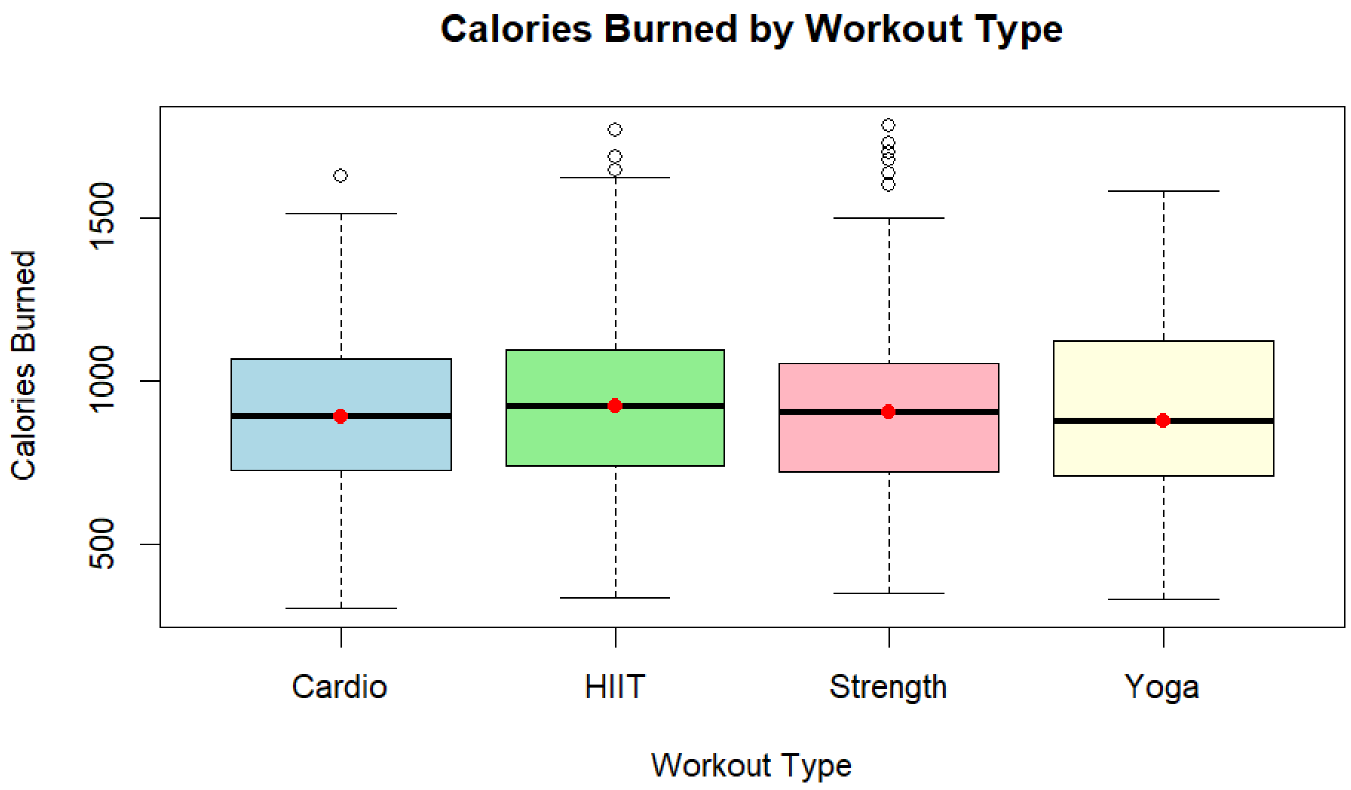
```{r}
# Kruskal-Wallis Test
kruskal_result <- kruskal.test(calories_burned ~ workout_type, data = data)
kruskal_result

```

```
        Kruskal-Wallis rank sum test

data:  calories_burned by workout_type
Kruskal-Wallis chi-squared = 1.4837, df = 3, p-value = 0.686
```

Since p-value is 0.686, larger than 0.05, we **failed to reject null hypothesis,** indicating there is **no significant difference** in average calories burned among the four workout types. This is somewhat surprising, as one might expect HIIT or Cardio to produce significantly higher calorie expenditure than Yoga.The lack of significant difference could reflect **variation in workout duration or intensity within each category**. For instance, a long and vigorous yoga session could burn as many calories as a short HIIT workout. This points to a **limitation of using broad exercise categories.** Stakeholders should track time and effort more precisely or consider hybrid programming to ensure energy expenditure targets are met.

The following box-plot visualize the comparison of BMI across workout frequencies:

```{r}
boxplot(calories_burned ~ workout_type, data = data,
        main = "Calories Burned by Workout Type",
        xlab = "Workout Type",
        ylab = "Calories Burned",
        col = c("lightblue", "lightgreen", "lightpink", "lightyellow"))


points(1:length(levels(data$workout_type)),
       tapply(data$calories_burned, data$workout_type, median),
       col = "red", pch = 19)

```



3.12 Examine whether there is a significant difference in the average calories burned by different types of exercise (Yoga, HIIT, Cardio, Strength) and gender. (Two -Way ANOVA)

This analysis explored whether **calories burned** differ significantly by **workout type**, **gender**, and the **interaction** between the two.

Independent Variable on workout type:

H0: All group means are equal across different workout type.

H1: At least one group mean is different across different workout type.

Independent Variable on Gender:

H0: All group means are equal between gender.

H1: There is significant difference in means between gender.

Interaction:

H0: There is no interaction effect between workout type and Gender on calories burned.

H1: There is an interaction effect between workout type and Gender on calories burned.

```{r}
#Q12
#Two-Way ANOVA
# Create a new group column (combined factors)
data$group <- interaction(data$workout_type, data$gender)

shapiro_results <- by(data$calories_burned, data$group, shapiro.test)
sapply(shapiro_results, function(x) x$p.value)

```

```
 Cardio.Female    HIIT.Female Strength.Female    Yoga.Female    Cardio.Male      HIIT.Male  Strength.Male      Yoga.Male
    0.19846132     0.75367312      0.55895173     0.18566327     0.06467408     0.11199693     0.06730510     0.02878872
```

```{r}
library(car)
leveneTest(calories_burned ~ workout_type * gender, data = data)

```

```
Levene's Test for Homogeneity of Variance (center = median)
       Df F value  Pr(>F)  
group   7  2.4719 0.01618 *
      965                  
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

Before running a two-way ANOVA, first checked 2 key assumptions:

1. Normality: The Shapiro-Wilk tests showed only one group out of 8 groups is **not normally distributed (Yoga.Male)** due to p-value < 0.05. While not ideal, two-way ANOVA is fairly robust with large samples.

2. Homogeneity of Variance: Levene’s Test indicated a **significant violation** (p = 0.016) due to p-value < 0.05, suggesting variance differences across groups.

Although one out of the eight Shapiro-Wilk tests indicated a deviation from normality, the remaining groups met the normality assumption. Given the large sample size (n = 974) and ANOVA's robustness to minor normality violations, we proceeded with Two-Way ANOVA. Besides, Levene's test indicated unequal variances . Therefore, the results should be interpreted with caution, because it violate both assumptions.

```{r}
# Ensure factor variables
data$gender <- as.factor(data$gender)
data$workout_type <- as.factor(data$workout_type)

# Two-Way ANOVA
two_way_result <- aov(calories_burned ~ workout_type * gender, data = data)
summary(two_way_result)

```

```
                     Df   Sum Sq Mean Sq F value   Pr(>F)    
workout_type          3   211670   70557   0.970    0.406    
gender                1  1635258 1635258  22.482 2.44e-06 ***
workout_type:gender   3   215615   71872   0.988    0.398    
Residuals           965 70189517   72735                     
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Factor	p-value	Significance
Workout Type	**0.406**	Not significant
Gender	$\mathbf{2.44 \times 10^{-6}}$	Highly significant
Interaction effect	**0.398**	Not significant

The **only significant factor** affecting calories burned was **gender**, with male members burning significantly more calories on average than female members. **Workout type** and the **interaction** between workout type and gender **were not significant.**

Additional Insight:

This finding may reflect physiological differences such as muscle mass and basal metabolic rate, which tend to be higher in males. From a gym management or health perspective, this result suggests that caloric expenditure programs may need to be gender-specific to reflect metabolic realities, especially if the goal is calorie-based performance tracking. This implies personalized plans by gender may yield better outcomes than adjusting by workout type alone. **Stakeholder should consider designing gender-sensitive fitness paths.**

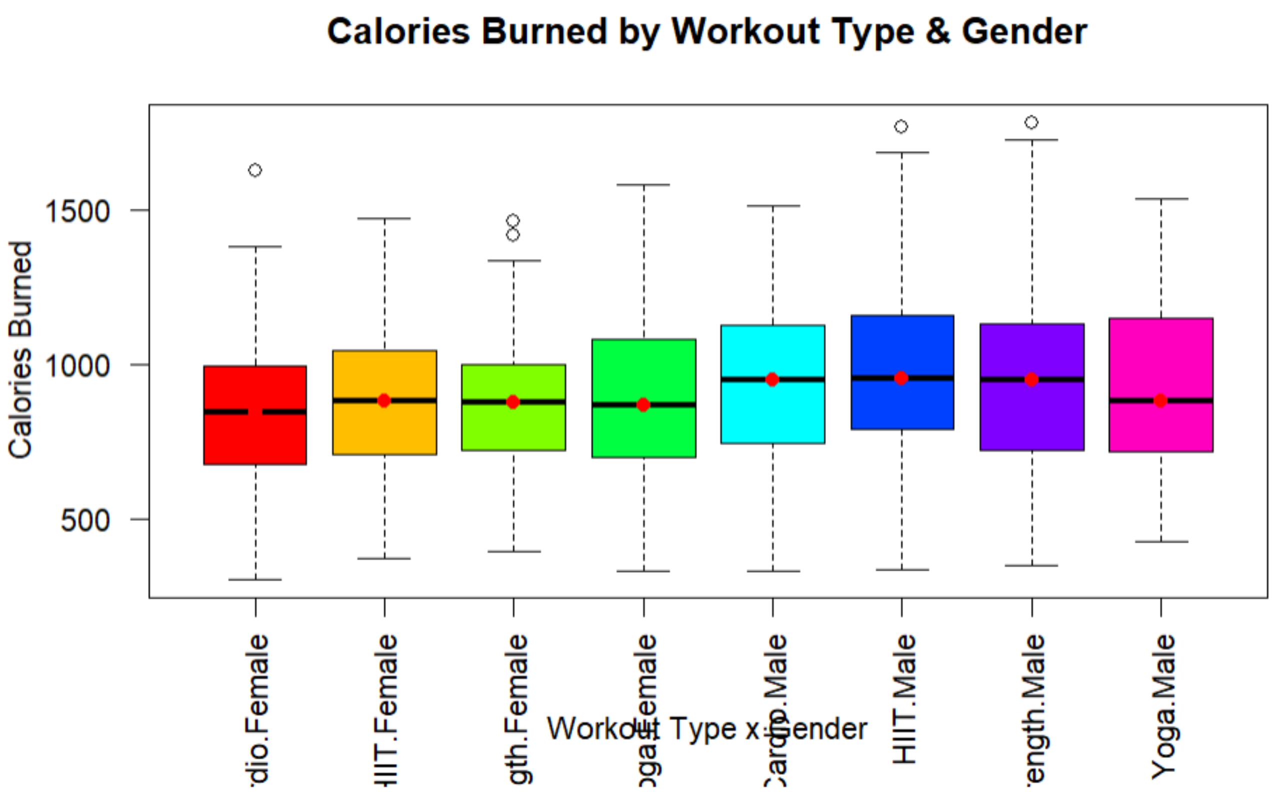
Interestingly, neither workout type nor its interaction with gender had a measurable effect. This suggests again that personalized workout intensity and duration, rather than just type, are likely to be the true drivers of caloric burn—a theme consistent across multiple analyses in this dataset.

The following box-plot visualize the comparison of BMI across workout frequencies:

````
```{r}
boxplot(calories_burned ~ group, data = data,
        main = "Calories Burned by Workout Type & Gender",
        xlab = "Workout Type x Gender", ylab = "Calories Burned",
        col = rainbow(length(unique(data$group))), las = 2)

# Add the median point of the group
points(1:length(unique(data$group)),
       tapply(data$calories_burned, data$group, median),
       col = "red", pch = 19)

```
````



3.13 How different factors affect calories burned? (Multiple Linear Regression Model)

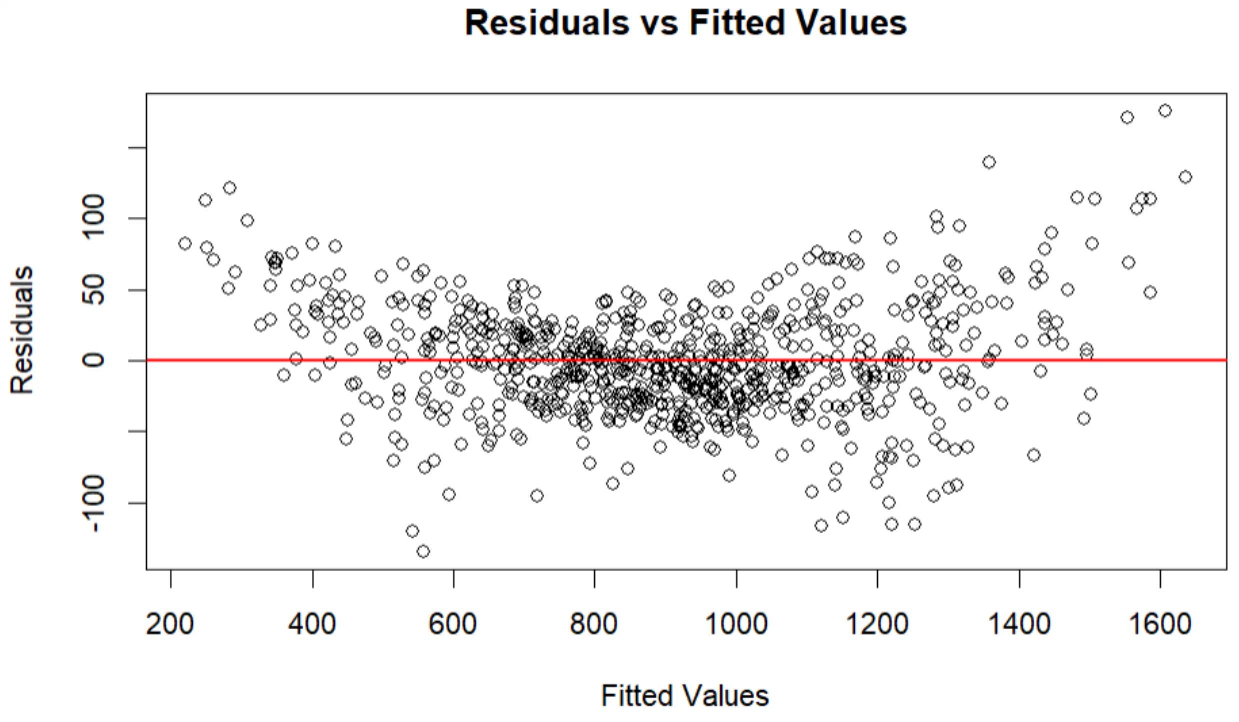
First, feature selection has been conducted, the author select almost all of the columns as regressors, excluding height, workout_frequency_days_week, experience_level, BMI, workout type, max_bpm and resting_bpm.Then a multiple regression model has been fit.

Discarded variables:

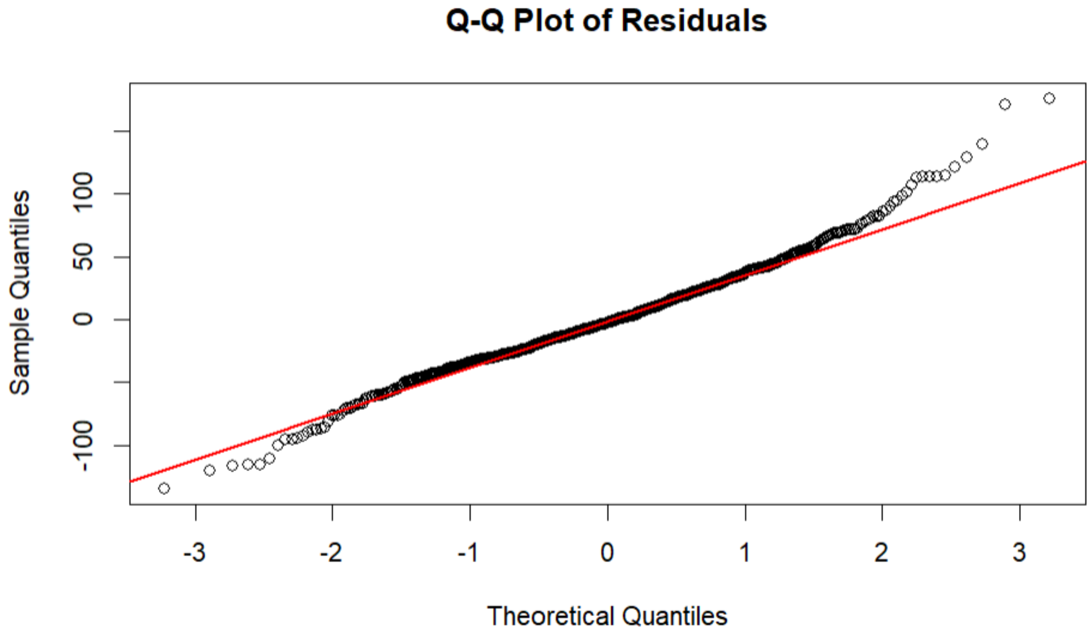
Variable	Reason
height	Usually weight can more directly reflect calorie consumption, height may be highly correlated with weight, causing multicollinearity.

workout_frequency_days_week	This is a long-term exercise habit, and has a low correlation with calories_burned in a single session.
experience_level	Similar to frequency, it is a long-term level; it has no direct relationship with the intensity of the current session.
BMI	Calculated from weight and height, weight is already included, adding BMI may cause redundancy
workout type	The author already proved that there is no significant difference in average calories burned among the four workout types in the previous problem statement.
max_bpm	avg_bpm is already a composite indicator of heart rate intensity and is a more stable input, max_bpm may be a more volatile measurement for outliers and is less reliable than the mean value
resting_bpm	Heart rate at rest before workout is clearly unrelated to calorie burned in common sense.

```
```{r}
#Q13
#Multiple Linear Regression Model

library(ggplot2)
library(car)          # for VIF
library(dplyr)

# Ensure gender is a factor
data$gender <- as.factor(data$gender)


# Build the multiple linear regression model
model <- lm(calories_burned ~ age + gender + weight_kg + avg_bpm +
              session_duration_hours + fat_percentage + water_intake_liters, data = data)

summary(model)
```

```
Call:
lm(formula = calories_burned ~ age + gender + weight_kg + avg_bpm +
    session_duration_hours + fat_percentage + water_intake_liters,
    data = data)

Residuals:
    Min      1Q  Median      3Q     Max
-132.39  -24.84   -1.84   23.22  176.07

Coefficients:
                        Estimate Std. Error t value Pr(>|t|)
(Intercept)           -795.62840   20.92300 -38.026   <2e-16 ***
age                     -3.41309    0.10487 -32.547   <2e-16 ***
genderMale              85.33670    4.06263  21.005   <2e-16 ***
weight_kg                0.09498    0.07375   1.288    0.198
avg_bpm                  6.25517    0.08875  70.480   <2e-16 ***
session_duration_hours 713.04331    4.88455 145.979   <2e-16 ***
fat_percentage          -0.41529    0.30472  -1.363    0.173
water_intake_liters     -1.35070    3.25197  -0.415    0.678
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 39.65 on 965 degrees of freedom
Multiple R-squared:  0.979,     Adjusted R-squared:  0.9789
F-statistic:  6428 on 7 and 965 DF,  p-value: < 2.2e-16
```
```

Individual t-test: (for every regressors)

H0: βj = 0 (The regressor xj is not important and hence can be removed from the model.)

H1: βj ≠ 0 (The regressor xj is important and hence should be remained in the model.)

Since the p-value of weight_kg, fat_percentage, and water_intake_liters are both more than 0.05, we **failed to reject null hypothesis,** indicating that they are not important and hence can be removed from the model. After remove them, we split the dataset into a training set (80%) and a testing set (20%) to evaluate the model's performance on unseen data, and fit a new model with taining set:

```
```{r}
set.seed(123)  # for reproducibility
#install.packages('caret')
library(caret)

# Split the data: 80% training, 20% testing
train_index <- createDataPartition(data$calories_burned, p = 0.8, list = FALSE)
train_data <- data[train_index, ]
test_data <- data[-train_index, ]

# Fit model on training data (using only significant variables)
model_reduced <- lm(calories_burned ~ age + gender + avg_bpm + session_duration_hours, data = train_data)
```
```

Then, model diagnostics has been performed with 4 steps:

```
```{r}
#Model Diagnostics
#Step 1: Residual Plot
plot(model_reduced$fitted.values, model_reduced$residuals,
     xlab = "Fitted Values",
     ylab = "Residuals",
     main = "Residuals vs Fitted Values")
abline(h = 0, col = "red", lwd = 2)


```
```



Step 1：The residuals vs fitted values plot is used to assess whether the residuals exhibit non-random patterns. Points appear to distribute quite evenly above and below the line residual = 0, implying that the linearity assumption is plausible. Also, the spread of the points does not seem to be very different for each fitted value, the homoscedasticity (constant variance) assumption is plausible as well. The assumptions of linearity and constant variance holds reasonably well.

````
```{r}
#Step 2: Q-Q Plot
# Q-Q Plot
qqnorm(model_reduced$residuals, main = "Q-Q Plot of Residuals")
qqline(model_reduced$residuals, col = "red", lwd = 2)

```
````



Step 2： The Q-Q plot checks whether residuals are approximately normally distributed. Most points lie close to the reference line, suggesting that the normality assumption is reasonably satisfied. This supports the validity of the inference from the regression model.

```
```{r}
#Step 3: Check for Multicollinearity using pairs scatter plot and correlation matrix

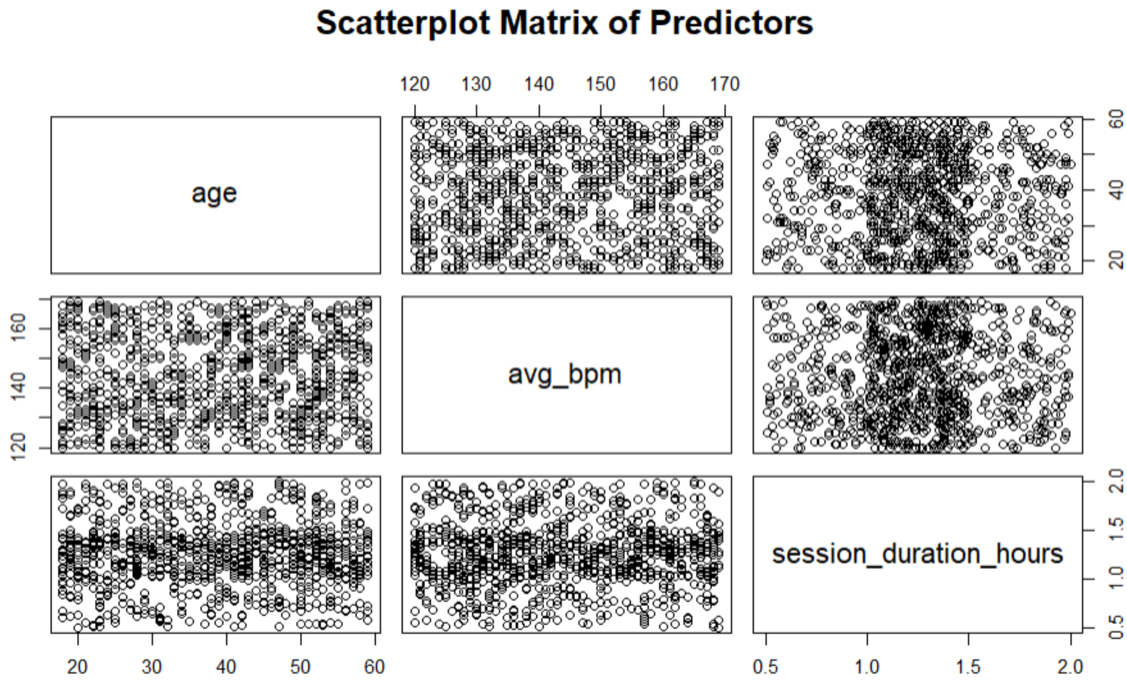
num_vars <- data[, c("age", "avg_bpm", "session_duration_hours")]

# 1. Pairwise scatter plot matrix
pairs(num_vars, main = "Scatterplot Matrix of Predictors")

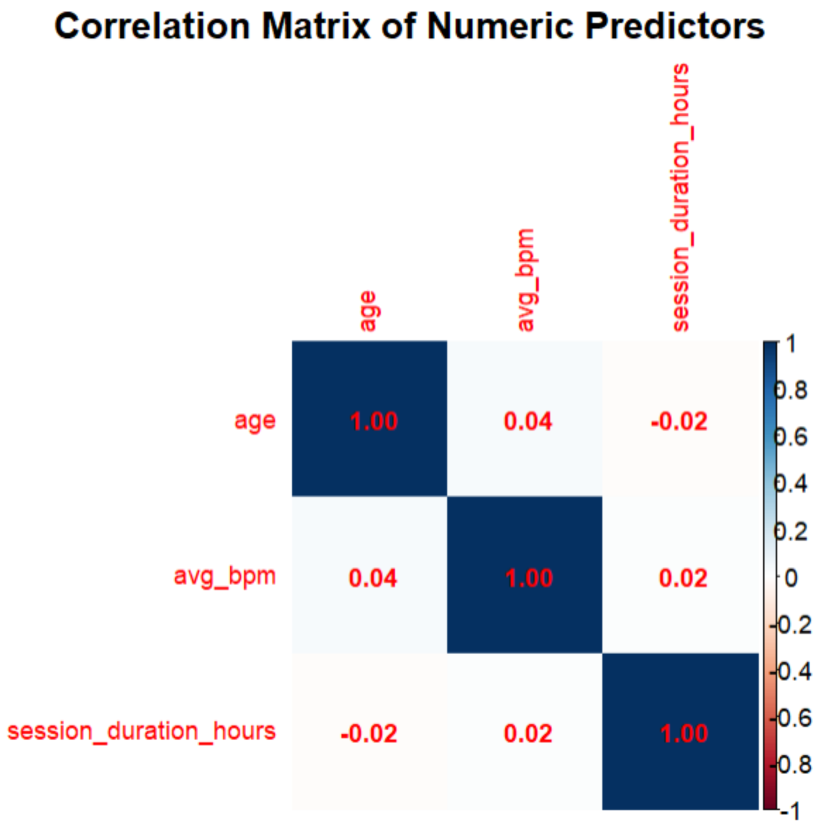
# 2. Correlation Matrix
cor_matrix <- cor(num_vars)

# install.packages("corrplot")
library(corrplot)

corrplot(cor_matrix, method = "color", addCoef.col = "red",
         tl.cex = 0.8, number.cex = 0.8,
         title = "Correlation Matrix of Numeric Predictors",
         mar = c(0,0,2,0))
```
```



Step 3：A **scatterplot matrix** was created to examine the pairwise linear relationships between the continuous independent variables (age, average heart rate, and session duration). The plots indicate **no strong linear correlations between predictors.** This suggests a low risk of multicollinearity.



The correlation matrix **further confirms that the variables are not highly correlated with each other.** All correlation coefficients are below the commonly accepted multicollinearity threshold of 0.8. Therefore, these variables are appropriate to be used together in the regression model without concerns of multicollinearity.

```{r}
#Step 3: Check for Multicollinearity using VIF
vif_values <- vif(model_reduced)
print(vif_values)

```

```
      age            gender            avg_bpm session_duration_hours
 1.002063          1.000786           1.002601               1.001945
```

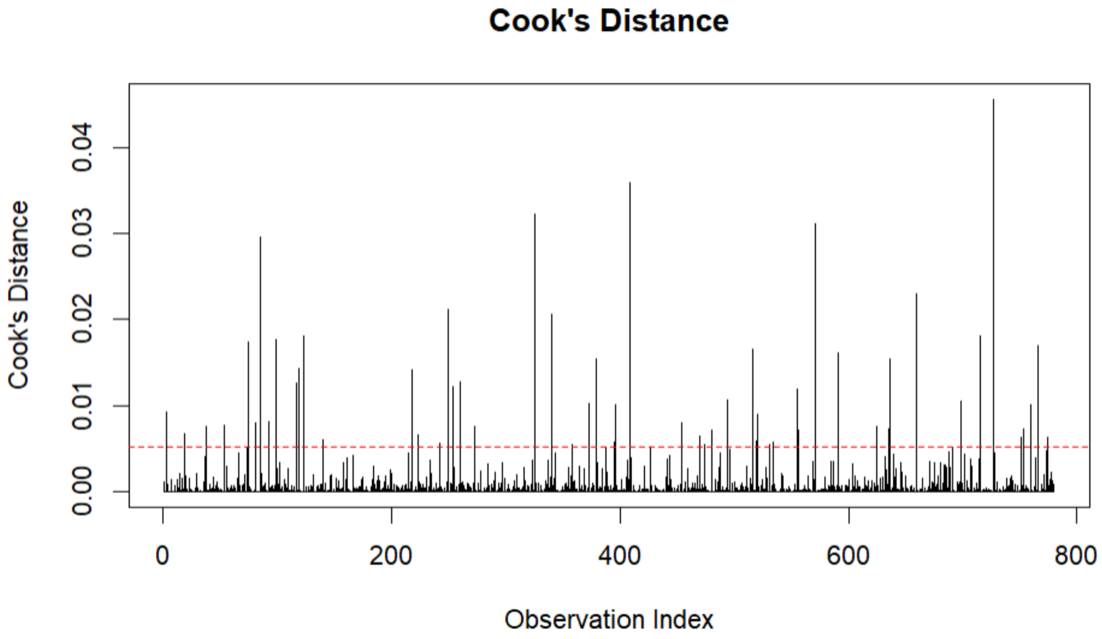
The Variance Inflation Factor (VIF) measures the degree of multicollinearity among the independent variables. In our model, all VIF values are below acceptable thresholds (<5), further confirming that **low correlation among multiple variables.** There is no multicollinearity among the independent variables.

```{r}
#Step 4: Cook's Distance
cooksD <- cooks.distance(model_reduced)

plot(cooksD, type = "h", main = "Cook's Distance",
     ylab = "Cook's Distance", xlab = "Observation Index")
abline(h = 4/length(cooksD), col = "red", lty = 2)

# Identify influential points
influential_points <- which(cooksD > 4 / length(cooksD))
print(influential_points)

```



Step 4： Cook's Distance is used to detect influential observations that may disproportionately affect the regression results. The red dashed line indicates the threshold (4/n), a commonly used rule-of-thumb.

Observations exceeding this threshold should be carefully reviewed.

From the plot, while several points exceed the threshold, their Cook's Distance values are still relatively low (all below 0.05), and no extreme influential points (Cook's D > 1) were found (Taneja, 2023). These mild influential observations are unlikely to distort the overall model fit. Therefore, the author decided to retain all observations in the dataset for the final model.

The model has met the basic linear regression modeling requirements after completing the four diagnostics

```{r}
summary(model_reduced)
```

```
Call:
lm(formula = calories_burned ~ age + gender + avg_bpm + session_duration_hours,
    data = train_data)

Residuals:
     Min       1Q   Median       3Q      Max 
-134.250  -26.078   -2.693   23.267  175.752 

Coefficients:
                        Estimate Std. Error t value Pr(>|t|)    
(Intercept)            -812.9217    15.6747  -51.86   <2e-16 ***
age                      -3.4897     0.1179  -29.60   <2e-16 ***
genderMale               88.9210     2.8375   31.34   <2e-16 ***
avg_bpm                   6.3087     0.0990   63.72   <2e-16 ***
session_duration_hours  716.6128     4.1711  171.81   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 39.57 on 775 degrees of freedom
Multiple R-squared:  0.9791,	Adjusted R-squared:  0.979 
F-statistic:  9058 on 4 and 775 DF,  p-value: < 2.2e-16
```

Since the p-value of all regressors coefficients are both very small, **reject null hypothesis,** indicating that they are important to the model. The fitted function is :

$$y = -812.9217 - 3.4897\,x1 + 88.9210\,x2 + 6.3087\,x3 + 716.6128\,x4$$

where x1 = age, x2 = genderMale, x3 = avg_bpm , x4 = session_duration_hour

Effect Size Interpretation:

- For every additional year of age, the calories burned decrease by 3.49 kcal.
- Men burn 88.9 kcal more than women.
- For every increase in heart rate (additional beat per second), the calories burned increase by 6.3 kcal.
- For every additional hour of exercise time, the calories burned increase by 716 kcal.

R-squared and Adjusted R-squared:

R-squared measures the proportion of variance in calories burned that can be explained by the model. In this model, $R^2 = 0.9791$, this means 97.91% of the variability in calories burned is explained by the model.

Adjusted R-squared accounts for the number of predictors and is a better measure for multiple regression. adjusted $R^2 = 0.979$, is almost same as the R^2. The model performs very well.

```
```{r}
# only select numeric variable
predictors <- data[, c("age", "avg_bpm", "session_duration_hours")]

scaled_predictors <- as.data.frame(scale(predictors))

scaled_predictors$gender <- data$gender
scaled_predictors$calories_burned <- data$calories_burned
```

```
```{r}
#fit model
model_scaled_reduced <- lm(calories_burned ~ age + gender + avg_bpm + session_duration_hours, data = scaled_predictors)
summary(model_scaled_reduced)
```
```

```
Call:
lm(formula = calories_burned ~ age + gender + avg_bpm + session_duration_hours,
    data = scaled_predictors)

Residuals:
    Min      1Q  Median      3Q     Max
-132.23  -25.41   -1.97   23.14  178.17

Coefficients:
                       Estimate Std. Error t value Pr(>|t|)
(Intercept)             858.815      1.846  465.35   <2e-16 ***
age                     -41.709      1.274  -32.75   <2e-16 ***
genderMale               88.746      2.547   34.84   <2e-16 ***
avg_bpm                  89.740      1.273   70.49   <2e-16 ***
session_duration_hours  245.870      1.273  193.21   <2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 39.66 on 968 degrees of freedom
Multiple R-squared:  0.9789,    Adjusted R-squared:  0.9788
F-statistic: 1.124e+04 on 4 and 968 DF,  p-value: < 2.2e-16
```

To evaluate the relative importance of each predictor, a standardized multiple regression model was fitted. Among all variables, **session duration had the largest standardized coefficient, indicating it is the most influential factor on calories burned**, followed by average heart rate and gender. Age showed the lowest impact on calories burned among the four regressors.
```

Besides, all model diagnostics were conducted on the original regression model. Since the standardized model only rescales the independent variables without altering the underlying residual structure or data points, repeating the diagnostics was deemed unnecessary.

```
```{r}
anova(model_scaled_reduced)
```
```

```
Analysis of Variance Table

Response: calories_burned
                         Df   Sum Sq  Mean Sq F value    Pr(>F)    
age                       1  1728668  1728668  1099.1 < 2.2e-16 ***
gender                    1  1733777  1733777  1102.4 < 2.2e-16 ***
avg_bpm                   1  8555701  8555701  5439.9 < 2.2e-16 ***
session_duration_hours    1 58711474 58711474 37330.0 < 2.2e-16 ***
Residuals               968  1522442     1573                      
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

Additionally, an ANOVA analysis was performed on the standardized model to further validate the relative importance of each predictor. The results were consistent: session duration had the largest F-value, followed by average heart rate,**indicating session duration is the most influential factor on calories burned.**

```
```{r}
# Predict on test set
predictions <- predict(model_reduced, newdata = test_data)

# Evaluate performance: RMSE and R-squared
rmse <- sqrt(mean((test_data$calories_burned - predictions)^2))
rsq <- cor(test_data$calories_burned, predictions)^2

cat("Test RMSE:", rmse, "\n")
cat("Test R-squared:", rsq, "\n")
```
```

```
Test RMSE: 40.11606
Test R-squared: 0.9784941
```

The fitted model on the training data was used to predict calorie burn on the testing data. The model achieved the following performance:

- Test R-squared: 0.9784941
- Test RMSE: 40.11606

This confirms that the model generalizes well and maintains high predictive accuracy outside the training data.

```{r}
new_data <- test_data[1:3, ]

# Confidence interval
conf_int <- predict(model_reduced, newdata = new_data, interval = "confidence")

# Prediction interval
pred_int <- predict(model_reduced, newdata = new_data, interval = "prediction")

cbind(new_data$calories_burned, conf_int, pred_int[, 2:3])
```

```
            fit       lwr       upr       lwr       upr
1 1313 1282.1123 1275.1032 1289.1214 1204.1199 1360.1048
2  883  910.7573  906.1414  915.3733  832.9435  988.5712
4  532  646.1799  637.6395  654.7203  568.0350  724.3248
```

Confidence Interval and prediction Interval has been calculated:

Confidence Interval (CI):

The CI represents the **uncertainty** in the **model's prediction of the "average calorie consumption value".**

For example, the predicted calorie consumption for the first observation is 1313 kcal, we are 95% confident that the average calorie consumption for "all users with the same input characteristics" falls between [1275, 1289] kcal.

The CI reflects the uncertainty in the estimation of the model coefficients.

Prediction Interval (PI) :

The PI represents the prediction range for **"a new individual"**, taking into account the residuals **(individual differences).**

For the same observation, the PI is [1204, 1360] kcal, which means that we are 95% confident that the actual calories burned by this new member in a single session will fall within this wider range.

PI is wider because it takes into account both model uncertainty and individual variation.

Conclusion of this problem statement:

In this study, we constructed a multiple linear regression model to predict calories burned based on several individual and session-specific factors. After evaluating the model diagnostics and removing non-significant predictors, we identified four key variables: age, gender, average heart rate (avg_bpm), and session duration. These variables were statistically significant and explained approximately 97.9% of the variation in calories burned (Adjusted $R^2 = 0.979$).

The final model passed all diagnostic checks, including residual analysis, normality Q-Q plot, multicollinearity (VIF < 5), and influence analysis (Cook's distance). This confirms the appropriateness and reliability of our regression model.

Additionally, we computed both confidence intervals (CI) and prediction intervals (PI) for selected test samples. The confidence intervals reflect the estimated range of average calorie burn for individuals with similar characteristics, while the wider prediction intervals reflect the variability in individual outcomes.

Overall, our findings suggest that **workout duration and average heart rate are the most impactful predictors of calories burned, followed by age and gender.** This means the best way to increase calories burned is to increase session time and manage workout intensity. Stakeholders can optimize programming and pricing by aligning with these drivers. Marketing messages should highlight how longer or more intense sessions yield measurable benefits.

4.0 Strategic Recommendations for Stakeholders

1. Increase Session Duration to Boost Calorie Burn

Since session duration is the most impactful predictor of calories burned, stakeholders should redesign class structures and personal training sessions to encourage slightly longer durations where feasible. Consider bundling 90-minute workouts as the default offering and create tiered membership benefits, for example: "long-session loyalty rewards" to motivate consistent members.

2.Monitor and Guide Workout Intensity via Heart Rate

The average heart rate is the second strongest predictor of calories burned. To leverage this, gym management can integrate wearable tech into group classes, for example, provide heart rate monitors, and train staff to guide members into target heart rate zones. This allows more individualized coaching and measurable feedback, which enhances member engagement and results.

3. Shift Focus from Workout Type to Session Quality

Workout type (HIIT, Yoga, Cardio, Strength) was not a significant factor in calorie burned. This suggests stakeholders should stop over-emphasizing "type" in program differentiation and instead focus on tracking session duration and intensity. Encourage instructors to clearly communicate calorie goals and trackable progress.

4. Promote Regular Participation Through Frequency-Based Rewards

Even though workout frequency did not significantly affect BMI or calorie burn in isolation, consistent attendance builds habits and long-term success. Stakeholders can launch engagement-driven programs like "burn streaks" or point-based challenges to reward weekly consistency, which indirectly supports better fitness outcomes.

5. Utilize Small-Scale Sampling for Fast Decision-Making

Our results show that even small samples (n = 30) can estimate population averages with high reliability. This empowers gym managers to conduct quick member surveys or test programs with small focus groups before large-scale implementation, saving both time and cost while maintaining confidence in insights.

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3. Barnes, T. (2024, January 16). Body fat percentage vs. BMI - Which is important? Baylor College of Medicine. https://www.bcm.edu/news/body-fat-percentage-vs-bmi-which-is-important

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APPENDIX 1

MARKING RUBRICS

Component Title	Assignment				Total Mark	100	
Criteria	Score and Descriptors					Weight (%)	Marks
	Excellent (9-8)	Good (7-5)	Average (4-3)	Need Improvement (2)	Poor (0-1)		
Section A	The source code fully addresses the question without compilation error. It has sensible variables.	The source code addresses the question without compilation error.	The source code partially addresses the question, but the program still works.	The source code partially addresses the question and has a compilation error.	The source code doesn't work and doesn't address the question. Or No Attempt	9	
	Excellent (14-11)	Good (10-8)	Average (7-5)	Need Improvement (4-3)	Poor (2-0)		
Section B	The source code fully addresses the question without compilation error. It has sensible variables.	The source code addresses the question without compilation error.	The source code partially addresses the question, but the program still works.	The source code partially addresses the question and has a compilation error.	The source code doesn't work and doesn't address the question. Or No Attempt	14	
	Excellent (23-19)	Good (18-14)	Average (13-8)	Need Improvement (7-4)	Poor (0-3)		
Section C	The source code fully addresses the question without compilation error. It has sensible variables.	The source code addresses the question without compilation error.	The source code partially addresses the question, but the program still works.	The source code partially addresses the question and has a compilation error.	The source code doesn't work and doesn't address the question. Or No Attempt	23	
	Excellent (24-19)	Good (18-14)	Average (13-8)	Need Improvement (7-4)	Poor (0-3)		
Section D	The source code fully addresses the question without compilation error. It has sensible variables.	The source code addresses the question without compilation error.	The source code partially addresses the question, but the program still works.	The source code partially addresses the question and has a compilation error.	The source code doesn't work and doesn't address the question. Or No Attempt	24	
					Subtotal	**70**	

	Excellent (30-24)	**Good (23-18)**	**Average (17-14)**	**Need Improvement (13-8)**	**Poor (7-0)**		
Report	The report meets all criteria exceptionally well.	The report meets most criteria well but has minor issues.	The report meets some criteria but has noticeable issues.	The report meets few criteria and has significant issues.	The report does not meet the criteria or is incomplete. Or No Attempt	**30**	
					Subtotal	**30**	
					TOTAL	**100**	

Note to students: Please include the marking rubric when submitting your coursework.